Refinancing Inequality and Monetary Transmission Channel

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Abstract

Mortgage refinancing is highly unequal across income groups, and this inequality substantially weakens the transmission of monetary policy. The inequality appears in both the intensity and timing of refinancing: the bottom income quintile refinances at only 60–65% of the top quintile's rate and faces markedly longer delays even when refinancing is financially beneficial. These gaps persist after controlling for lender-side credit tightness, borrower credit quality, and origination lender selection. The associated unrealized savings are large, exceeding 7.6% of monthly income for the bottom quintile. I develop a structural mortgage refinancing model that decomposes refinancing frictions into two distinct channels — inattention and hassle cost — and allows both to vary by income state. A homogeneous-friction benchmark fails to reproduce the observed income gradient; income-state-dependent frictions are necessary to match the data. A counterfactual intervention that raises refinancing attention by the magnitude documented in field evidence increases the five-year cumulative consumption response to a policy rate cut by approximately 10%, with roughly 80% of the additional response accruing to the bottom two income quintiles. This concentration reflects the joint distribution of refinancing frictions and marginal propensities to consume, and implies that attention-targeted interventions can simultaneously strengthen monetary transmission and compress liquid-wealth inequality.

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1 Introduction

Mortgage refinancing is one of the largest discretionary cash-flow decisions a U.S. household can make. Nearly 96% of outstanding U.S. residential mortgages carry a fixed rate, so when the Federal Reserve cuts the policy rate, the primary pathway for cash-flow relief to reach household balance sheets runs through refinancing rather than automatic payment resets. During the 2020–2021 refinancing boom, the New York Fed estimates that 14 million mortgages were refinanced; rate refinancers reduced monthly payments by $220 on average, and roughly 9 million non-cash-out refinancers lowered annual housing costs by $24 billion.

A large literature documents that households nevertheless under-refinance even when refinancing is financially valuable, attributing this inaction to information frictions, inattention, fixed costs, and behavioral biases (Keys et al. 2016; Agarwal et al. 2016; Andersen et al. 2020; Zhang 2022). A closely related strand quantifies how these refinancing frictions attenuate the cash-flow channel of monetary policy in the aggregate (Beraja et al. 2019; Defusco and Mondragon 2020; Berger et al. 2021; Eichenbaum et al. 2022; Byrne et al. 2023). Existing work that documents borrower-level heterogeneity has done so primarily along demographics, race, and behavioral correlates (Andersen et al. 2020; Gerardi et al. 2023; Agarwal et al. 2024).

Despite this progress, the literature has paid relatively limited attention to a dimension that, on first principles, ought to matter most for aggregate consumption: how refinancing frictions are distributed across the income distribution, and how that distribution shapes the strength of monetary policy transmission. Income is tightly linked to liquidity constraints and to the marginal propensity to consume out of cash-flow relief. If refinancing systematically bypasses higher-MPC households at the bottom of the income distribution, the aggregate consumption response to a rate cut is mechanically dampened, and structural models with homogeneous refinancing frictions cannot capture this attenuation.

Figure 1 previews the central empirical pattern. Plotting prepayment activity alongside mortgage rates by income quintile, the figure shows that prepayment rises sharply when interest rates fall — consistent with refinancing-driven activity — but the magnitude of the response differs

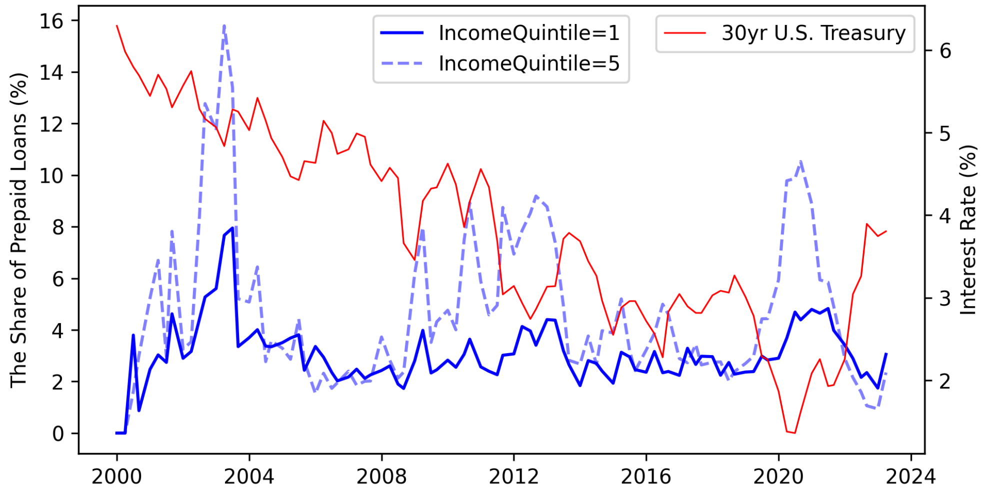


Figure 1: Prepayment Activity by Income Quintile

Notes: The figure plots refinancing activity by income quintile using a 10 percent sample of the Freddie Mac loan-level performance data merged to origination records. Refinancing activity is measured quarterly as the share of outstanding loans in each income quintile that voluntarily prepaid during the quarter. The figure reports income quintiles 1 and 5 for readability. The right axis plots the quarterly average 30-year U.S. Treasury yield from FRED.

substantially across income groups facing the same aggregate interest-rate environment. Higher-income borrowers refinance much more aggressively than lower-income borrowers. Following Agarwal et al. (2024), I refer to this systematic income gradient in refinancing behavior as *refinancing inequality*, and study it along two dimensions: the *intensity* of refinancing (the conditional probability of refinancing once a loan becomes financially advantageous to refinance) and its *timing* (how long borrowers wait to act once refinancing becomes profitable).

The central empirical challenge is that income at origination is correlated with factors that could independently generate the observed refinancing gap, including credit scores, home equity, loan size, geographic location, lender selection, and program eligibility. A related concern is that the observed income gradient may reflect time-varying lender-side credit-supply tightening that disproportionately affects lower-income borrowers rather than borrower-side refinancing frictions. I address these challenges using a loan-level panel from the Freddie Mac Single-Family Loan-Level Dataset covering loans originated between 2000 and 2022. I classify each loan-quarter as financially advantageous to refinance using both a simple rate-gap measure and the optimal refinancing threshold of Agarwal et al. (2013), and control throughout for quarterly updated loan-to-value ratios constructed from Zillow house price indices, a Federal Reserve Senior Loan Officer Survey credit-tightness index, Home Mortgage Disclosure Act (HMDA)-based income-quintile-specific mortgage denial rates, and credit-score quartile stratification.

I document three main facts. First, refinancing inequality is large in intensity: once households become financially advantageous to refinance, the bottom income quintile refinances at only 60-65% of the top quintile's rate, and the gap is most pronounced precisely in the region where the financial case for refinancing is strongest. Second, the inequality also appears in *timing*, a dimension that has received limited attention in the literature. Aligning loans at the month they first become financially advantageous to refinance, I find that the top income quintile reaches an 80% refinancing survival rate after roughly four months, whereas the bottom income quintile takes approximately ten months. Third, the inequality is economically meaningful at the household level. By matching pre- and post-refinancing loans, I estimate that households in the bottom income quintile leave unrealized refinancing savings exceeding 7.6% of monthly income — a persistent and recurring cash-flow shortfall over the remaining life of the loan.

Two residual identification concerns — whether the gradient reflects sorting into origination lenders with systematically different long-run servicing practices, and whether income quintile at origination diverges from that at the refinancing decision — are addressed directly. To test for origination-lender sorting, I merge the Freddie Mac panel to HMDA records and include origination lender-by-year fixed effects that compare refinancing behavior across income groups within the same lender-vintage cell; the income gradient remains economically large and statistically significant. To address potential income mismeasurement, I re-estimate the baseline specifications using income quintiles reassigned at the refinancing date from matched refinancing-loan records, restrict the sample to borrowers whose income quintile remains unchanged between origination and refinancing, and use the PSID to bound cross-quintile mobility over a two-year horizon. The refinancing gap remains robust across all three approaches.

Beyond these cross-sectional findings, I provide what is, to my knowledge, the first direct evidence on how the dynamic refinancing response to identified monetary policy shocks varies by fundamental borrower characteristics. Prior work documents the aggregate dynamic response

(Eichenbaum et al. 2022) and heterogeneity driven by loan-state characteristics or regional housing equity distributions (Beraja et al. 2019). How the dynamic response varies by who the borrower is — especially by income, the dimension that simultaneously determines marginal propensity to consume — has not been studied. Using a regional Freddie Mac panel and an LP-IV design based on the high-frequency monetary surprises of Swanson (2021), I find that the differential cumulative refinancing response between the top and bottom income quintiles peaks at approximately 4.3 percentage points two quarters after a 100-basis-point easing. This magnitude is substantial relative to the average aggregate refinancing response of about 2.7 percentage points at the same horizon, underscoring that the income gradient is economically meaningful.

To interpret these empirical patterns and assess their aggregate implications, I develop a structural mortgage refinancing model that nests Berger et al. (2021) but allows two distinct refinancing frictions — an attention margin and an execution-cost margin — to vary across income states. Two features of the model are central. First, a calibrated homogeneous-friction benchmark based on the Berger et al. (2021) specification cannot account for the observed income gradient, instead generating near-uniform refinancing rates across income quintiles where the data exhibit a steep and monotonic increase. Matching the observed gradient therefore requires income-state-dependent refinancing frictions; variation in the average level of frictions alone is insufficient. Second, the model decomposes the observed refinancing gradient into attention and execution-cost components and finds that, consistent with the broader evidence in Berger et al. (2021), the attention channel is the dominant driver of refinancing inequality across the income distribution.

I then use the calibrated model to quantify how refinancing frictions shape the transmission of monetary policy. Based on the field-experiment evidence in Byrne et al. (2023), my main counterfactual considers a market-wide notice intervention that raises the attention-capacity parameter by 60 percent across all income states. Under this evidence-based attention counterfactual, the model-implied five-year discounted cumulative consumption response to a 100 basis-point policy rate cut rises from 3.13 percentage points in the baseline economy to 3.44 percentage points, corresponding to an amplification of approximately 10.0 percent. The additional response is highly concentrated among lower-income households: the bottom two income quintiles account for roughly 80 percent of the discounted consumption gain. This concentration reflects the joint distribution of refinancing gains, attention frictions, and marginal propensities to consume, and demonstrates that even a market-wide reduction in refinancing frictions transmits disproportionately through households for whom refinancing-induced cash-flow relief has the largest spending effects.

Related Literature This paper contributes to three strands of literature. The first studies refinancing frictions and the refinancing channel of monetary policy. A large body of work documents that refinancing frictions attenuate the cash-flow pass-through of rate cuts (Keys et al. 2016; Agarwal et al. 2016; Di Maggio et al. 2017), and that this attenuation is state-dependent and history-dependent through housing equity and past interest-rate paths (Beraja et al. 2019; Wong 2021; Berger et al. 2021; Eichenbaum et al. 2022). Byrne et al. (2023) identifies inattention as a causal driver of incomplete pass-through. Relative to this strand, my contribution is twofold. First, I show that not only the average level of refinancing frictions but also their *distribution* across income groups matters for monetary transmission: because cash-flow relief systematically bypasses lower-income households with higher marginal propensities to consume, the aggregate consumption response is dampened in a way that average-friction models cannot capture. Second, I document a new empirical regularity: the dynamic refinancing response to identified monetary policy shocks varies substantially by borrower income — a dimension prior work left uncharacterized. Third, the structural decomposition separates inattention from execution costs and identifies the former as the dominant driver of the income gradient, providing a clean target for policy intervention.

The second strand examines heterogeneity in refinancing behavior across borrower characteristics (Andersen et al. 2020; Gerardi et al. 2023) and its redistributive implications (Berger et al. 2024; Fisher et al. 2024; Zhang 2022). This paper is closely related to contemporaneous reduced-form evidence by Jørring and Westphal (2025), which uses survey data to document that refinancing take-up is systematically lower among higher-MPC households, and to Agarwal et al. (2024), which documents income-heterogeneous refinancing behavior during the COVID-19 episode. My work complements theirs while substantially broadening both the empirical scope and the quantitative framework: I document refinancing inequality not only in intensity but also in timing using a 22-year administrative loan-level panel, establish that the gradient survives controls for lender-side credit tightness and origination lender selection, and embed income-state-dependent refinancing frictions in a structural model that quantifies counterfactual policy gains unavailable to reduced-form approaches.

The third strand connects monetary policy to inequality (Coibion et al. 2017; Kaplan et al. 2018; Auclert 2019; Amberg et al. 2022; Andersen et al. 2023). My contribution to this literature is to provide a refinancing-based microfoundation for how accommodative monetary policy interacts with income inequality through unequal cash-flow pass-through, and to show — through the model counterfactuals — that interventions that compress income-specific refinancing frictions could simultaneously generate a visible compression of liquid-wealth inequality.

The remainder of the paper proceeds as follows. Section 2 describes the data and key measurement choices. Section 3 documents the empirical facts on refinancing inequality in intensity and timing, addresses identification threats, and estimates the unrealized savings. Section 4 develops the structural model and the homogeneous-friction benchmark comparison. Section 5 presents the calibration, untargeted moments, and the monetary policy counterfactual. Section 6 concludes.

2 Institutional Background and Data

2.1 Effect of Income on Refinancing

The potential mechanisms through which income may influence refinancing behavior can be grouped into three broad channels. First, lower-income households may face refinancing frictions arising from limited liquidity and the presence of closing costs. Closing costs typically include fees for loan applications, appraisals, title insurance, and other administrative services, and generally amount to 2–5% of the loan balance. These costs can constitute a substantial barrier to refinancing for households with limited liquid assets. Although borrowers can often roll closing costs into the new loan balance, doing so typically involves a trade-off with the refinancing interest rate. Specifically, borrowers may choose between paying higher upfront costs in exchange for a lower interest rate — thereby reducing future monthly payments — or paying lower upfront costs while accepting a higher interest rate, which increases future monthly payments and total interest expenses. Appendix Figure A.1 illustrates this trade-off between upfront refinancing costs and mortgage rates. In the example shown, both cash-out and non-cash-out refinancing options are associated with lower interest rates when borrowers incur higher upfront costs.

Second, refinancing behavior may be influenced by payment-to-income (PTI) constraints embedded in mortgage underwriting standards. In the United States, PTI ratios are commonly incorporated into the loan approval process as soft borrowing constraints. For example, the standard PTI threshold for conventional mortgages is typically around 36%, while higher PTI ratios are generally approved only when borrowers satisfy compensating conditions such as low loan-to-value ratios or high credit scores. As shown in Table A.7, lower-income households exhibit systematically higher PTI ratios at origination than higher-income households. Consequently, given the same distribution of idiosyncratic income shocks, lower-income households are more likely to encounter PTI-related refinancing constraints.

Third, income may be associated with differences in borrower attention and refinancing awareness. Andersen et al. (2020) show that inattention is an important source of suboptimal refinancing behavior and is systematically correlated with household characteristics associated with financial sophistication. As a result, behavioral frictions related to limited attention or delayed action may generate systematic differences in refinancing behavior across income groups.

2.2 Data Overview

The primary dataset used in this study is the Freddie Mac Single Family Loan-Level Dataset, which covers loans originating from 2000 to 2022. This dataset provides comprehensive information on both loan issuance and performance.[1] The issuance data includes a rich set of loan and borrower characteristics at the time of origination. Key variables include the initial loan balance, loan term, interest rate, credit score, Loan-to-Value (LTV) ratio, Payment-to-Income (PTI) ratio, ZIP code, and estimated income. These variables allow for detailed analysis while controlling for factors influencing refinancing decisions, and I restricted sample to fixed-rate mortgages. In addition

[1]The Freddie Mac data set represents approximately 20%-30% of the total U.S. mortgage market, with the notable exclusion of "jumbo" loans, which are larger than the conforming loan limits set by the Federal Housing Finance Agency every year.

to issuance data, the performance data offers monthly observations of loans, which include the remaining balance, current interest rate, amortization, and prepayment status. This panel data enables the tracking of loan performance over time, facilitating a dynamic analysis of refinancing behavior.

The income for each loan can be back-calculated using Freddie Mac's issuance data, using the original interest rate, loan amount, maturity, and PTI information. Based on the distribution of this derived income of each origination year, the loans were then divided into quintiles.

To complement the Freddie Mac data, this study also utilizes data from the Home Mortgage Disclosure Act (HMDA) from 2000 to 2022.[2] The HMDA data consists of detailed records on individual mortgage applications and originations collected by financial institutions. The data include a wide array of information, such as the loan amount, type of loan, property location, borrower characteristics, and the outcome of the application. I matched Freddie Mac to HMDA using common variables to find corresponding loans, and used information from HMDA for analysis.

Detailed summary statistics of the final panel used in the analysis can be found in Appendix A.1.

2.3 Identifying Refinancing Loans

A central empirical challenge in identifying refinancing activity from the Freddie Mac loan panel is that the dataset reports mortgage prepayments but does not provide the reason for prepayment. Because mortgage prepayments may occur for reasons other than refinancing — such as household moves, home sales, or property trading — accurately identifying refinancing events is a critical step in the analysis. Following Agarwal et al. (2024), I identify refinancing activity by matching a prepaid loan to a newly originated Freddie Mac-funded loan within a 45-day window surrounding the termination date of the prepaid loan.[3]

[2]The Home Mortgage Disclosure Act (HMDA) requires financial institutions to maintain and annually disclose data about home purchases, home purchase pre-approvals, home improvement, and refinance applications. The HMDA data captures the majority of mortgages in the U.S., accounting for about 92 percent of originations nationally in 2017, according to the Consumer Financial Protection Bureau.

[3]The 45-day window is motivated by the institutional structure of mortgage rate locks. A rate lock is an agreement between the borrower and the lender that guarantees a specified mortgage rate for a predetermined period while the mortgage application is processed. A 45-day lock period is common in practice. Consequently, when a loan is prepaid for refinancing purposes, the replacement loan is highly likely to be originated within this interval.

A further challenge arises from the limited geographic granularity in the publicly available Freddie Mac data. Specifically, the dataset reports only three-digit ZIP codes, which correspond to relatively large geographic areas and generate numerous duplicate matches, making unique loan identification infeasible. To address this issue, I match the Freddie Mac data to HMDA records, exploiting the finer geographic identifiers available at the census-tract level in HMDA. Because the matching procedure may itself introduce selection concerns or matching errors, I conduct a series of validation exercises reported in Appendix A.2. These tests indicate that the matched sample closely resembles the underlying population and does not exhibit evidence of economically meaningful matching bias.

2.4 Identifying In-the-money Loans

The decision for households to refinance is determined by two motives: first, the cash-out motive due to liquidity needs in response to idiosyncratic shocks, and second, the demand for reducing financing costs through lower mortgage rates. The cash-out motive is known to be insensitive to interest rates, while the interest-rate reduction motive is directly affected by monetary policy.[4] Since the focus of this study is to analyze the differences in the transmission of the benefits of accommodative monetary policy across income groups, the interest-rate reduction motive should be emphasized. Therefore, distinguishing whether there is a financial benefit from lowering mortgage rates during refinancing is one of the key identifications. An in-the-money option refers to an option that provides a financial benefit when exercised. Therefore, a fixed-rate mortgage becomes an in-the-money loan when refinancing is financially advantageous.

$$\text{In-the-money} \;\; if \;\; c_i - \tilde{m}_{it} > \theta_{it} \tag{1}$$

The financial benefit of refinancing arises when the borrower's current mortgage rate exceeds the rate available on a newly originated loan, that is, when the loan is "in-the-money." Formally, a loan is identified as in-the-money when the existing coupon c_i is higher than the prevailing

[4] Chen et al. (2020) focused on the cash-out motive of refinancing and suggested that refinancing decisions of low-liquidity households are insensitive to interest rate changes.

obtainable rate $\tilde{m}_{it}$ by more than the threshold θ_{it}, as shown in Equation 1. The key object in this condition is the rate gap, defined as $g_{it} \equiv c_i - \tilde{m}_{it}$: the gap between the borrower's current coupon and the rate they could obtain today. To construct a clean measure of this gap, I estimate the obtainable rate $\tilde{m}_{it}$ as the median origination rate among newly originated loans within a granular borrower–loan characteristic cell defined by origination quarter, ZIP code, credit score bucket, loan-to-value ratio, and income quintile, following Defusco and Mondragon (2020). This cell-based approach compares each borrower to observationally similar borrowers who actually refinanced in the same local market at the same point in time, yielding a benchmark rate that reflects what the borrower could plausibly obtain given their characteristics. Crucially, this measure of the rate gap is held fixed when comparing refinancing behavior across income groups, so that any differences in take-up rates across the income distribution reflect heterogeneity in the response to a given gap rather than differences in the gap itself.

In the baseline specification, I set $\theta_{it} = 0$, assuming that refinancing is financially beneficial whenever the contracted rate exceeds the estimated market rate. As a robustness check, I also employ a more conservative identification rule based on the Agarwal-Driscoll-Laibson adjusted rate gap (hereafter *ADL-adjusted rate gap*), where $\theta_{i,t} = \theta_{it}^{ADL}$ corresponds to the optimal refinancing threshold derived from Agarwal et al. (2013). This threshold incorporates refinancing costs, expected exogenous prepayment, remaining maturity, and interest rate volatility, thereby tightening the condition under which a loan qualifies as in-the-money. Although more restrictive, this ADL-based rule yields qualitatively similar patterns of refinancing inequality. The detailed formula of the ADL-adjusted rate gap is described in Appendix B.

3 Evidence of Refinancing Inequality

3.1 Average Refinancing Probability

To quantify the income gradient in refinancing intensity, I estimate a linear probability model (LPM) of monthly refinancing, restricting the sample to loan-months in which the loan is financially advantageous to refinance.[5] Restricting attention to in-the-money loan-months focuses the analysis on whether households act on financially profitable refinancing opportunities, abstracting from refinancing driven by idiosyncratic liquidity needs such as equity extraction.[6]

$$Refi\_Indicator_{it} = \alpha + \sum_{q=2}^{5} \beta_q \cdot \mathbf{1}[IncomeQuintile_i = q] + \Gamma' X_{it} + \nu_{z(i)} + \lambda_{v(i)} + \epsilon_{it} \tag{2}$$

The dependent variable *Refi_Indicator*$_{it}$ equals one if loan i refinances in month t and zero otherwise. The coefficients β_q measure the refinancing propensity of income quintile q relative to the bottom quintile (Q1), conditional on being in-the-money. X_{it} is a vector of borrower and loan controls, $\nu_{z(i)}$ and $\lambda_{v(i)}$ denote ZIP-code fixed effects and origination vintage year-quarter fixed effects respectively.

Cross-quintile differences in refinancing propensity could reflect differences in observable borrower and loan characteristics rather than an independent income effect. I therefore include a rich set of controls — credit score, loan age, loan balance-to-income ratio, and loan balance, each entered as categorical variables to allow flexible nonlinear effects — along with ZIP-code and origination vintage year-quarter fixed effects. Two time-varying threats require more targeted treatment beyond these standard controls.

Equity distribution A first source of potential confounding is the equity channel. As Beraja et al. (2019) emphasize, lower-income borrowers tend to purchase homes in areas with greater house price cyclicality, increasing their exposure to episodes of negative equity that prevent refinancing. Regional fixed effects cannot absorb this time-varying channel. I address it by constructing a monthly LTV estimate for each loan, combining origination LTV and the current outstanding balance with the Zillow ZIP code-level house price index to track collateral values over time. I additionally control directly for participation in the Home Affordable Refinance Program (HARP),

[5] Results are robust to estimation using a logit specification; because I use categorical income-quintile indicators, the linear probability model imposes no additional functional-form restriction on the relationship of interest.

[6] For example, households facing idiosyncratic income shocks may refinance to extract home equity regardless of the rate environment, as documented by Chen et al. (2020). Including such refinancings would conflate distinct behavioral margins.

which allowed underwater borrowers to refinance from 2009 to 2018 and is mechanically tied to the equity channel.

Lenders' credit tightness A second concern is that lenders may tighten credit supply differentially across income groups during downturns, so that the observed refinancing gap partly reflects supply-side rationing rather than a borrower-side friction. I address this directly by including two controls: quarterly changes in the Federal Reserve Senior Loan Officer Survey (SLOOS) credit-tightness index, and HMDA-based income-quintile-specific mortgage denial rates constructed using the same quintile boundaries as the main loan panel.[7]

Table 1: Average Refinancing Rate by Income Quintile (Simple Rate Gap)

Dependent	Refinancing Indicator × 100			
Equation	(1)	(2)	(3)	(4)
IncomeQuintile 2	0.213^a	0.316^a	0.324^a	0.296^a
	(0.133)	(0.115)	(0.117)	(0.131)
IncomeQuintile 3	0.072^a	0.070^a	0.077^a	0.080^a
	(0.115)	(0.117)	(0.117)	(0.131)
IncomeQuintile 4	0.809^a	0.647^a	0.649^a	0.470^a
	(0.115)	(0.117)	(0.117)	(0.131)
IncomeQuintile 5	1.452^a	1.573^a	1.672^a	1.153^a
	(0.133)	(0.115)	(0.117)	(0.131)
Constant	1.923^a	0.421^a	0.504^a	0.562^a
	(0.064)	(0.199)	(0.218)	(0.215)
Control variables				
LTV, credit, loan age, rate inc., ZIP		x	x	x
Credit tightness, HARP			x	x
Loan balance, Est. LTV				x
Observations	1,342,423	1,342,423	1,342,423	1,342,423
R-squared	0.002	0.010	0.011	0.011

Notes: Superscript a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis. Standard errors are clustered by zipcode and original vintage year.

Table 1 reports estimates under the simple rate-gap definition ($\theta_{it} = 0$). The unconditional specification in column (1) yields a top-to-bottom income-quintile gap in monthly refinancing probability of 1.45 percentage points. As controls are progressively added, the gap narrows to 1.15 percentage points in the full-control specification in column (4). In predicted-probability terms, the full-control specification implies a monthly refinancing rate of 2.00 percent for the bottom quintile and 3.10 percent for the top quintile: bottom-quintile borrowers refinance roughly 35 percent less

[7]The latter measure captures differential credit availability specifically for each income quintile, allowing the specification to absorb time-varying, income-specific supply-side variation.

frequently than top-quintile borrowers, conditional on facing an equivalent rate incentive. Figure 2 plots these predicted probabilities across all specifications, illustrating that the income gradient is robust to the progressive addition of controls and that the ordering across quintiles is strictly monotone throughout. Detailed predicted refinancing rates by income quintile are reported in Table A.10.

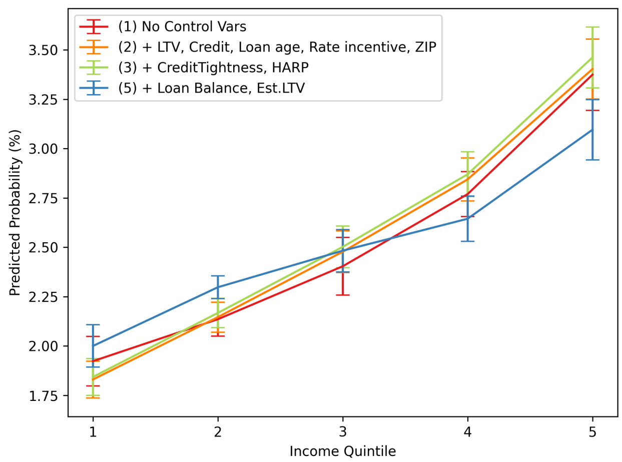


Figure 2: Predicted Refinancing Probability by Income Quintile

Notes: This plot shows the predicted monthly refinancing probability conditional on being in-the-money, using Equation 2. The 95% confidence intervals are represented by error bars and each line corresponds to the columns of the regression result in Table 1. Standard errors are clustered by ZIP code and original vintage year.

Table 2 re-estimates the specification using the ADL-adjusted rate gap, which applies the more conservative in-the-money criterion of Agarwal et al. (2013). Under this stricter definition, all income-quintile coefficients remain positive, monotone, and statistically significant at the 1% level. In the full-control specification, top-to-bottom income-quintile gap is 1.57 percentage points. The predicted monthly refinancing probability rises from 2.13 percent for the bottom quintile to 3.44 percent for the top quintile, a gap of 1.31 percentage points as shown in Table A.11; bottom-quintile borrowers refinance approximately 38 percent less frequently than top-quintile borrowers. The qualitative patterns are thus robust across both definitions of the refinancing incentive, and the gradient is if anything more precisely estimated under the ADL criterion.

A remaining concern is that the income gradient could reflect unobserved aggregate conditions that vary over time, rather than differences in refinancing behavior across borrowers. As an

Table 2: Average Refinancing Rate by Income Quintile (ADL-Adjusted Rate Gap)

Dependent	Refinancing Indicator × 100			
Equation	(1)	(2)	(3)	(4)
Income Quintile 2	0.518[a]	0.562[a]	0.577[a]	0.579[a]
	(0.065)	(0.068)	(0.075)	(0.072)
Income Quintile 3	0.924[a]	1.030[a]	1.064[a]	0.923[a]
	(0.093)	(0.107)	(0.113)	(0.100)
Income Quintile 4	1.321[a]	1.472[a]	1.501[a]	1.148[a]
	(0.120)	(0.125)	(0.131)	(0.118)
Income Quintile 5	1.872[a]	1.909[a]	2.057[a]	1.571[a]
	(0.139)	(0.140)	(0.151)	(0.135)
Constant	1.940[a]	1.199[a]	1.317[a]	1.05[a]
	(0.064)	(0.199)	(0.218)	(0.215)
Control variables				
LTV, credit, loan age, rate inc., ZIP		x	x	x
Credit tightness, HARP			x	x
Loan balance, Est. LTV				x
Observations	1,342,278	1,342,278	1,342,278	1,342,278
R-squared	0.001	0.013	0.014	0.014

Notes: Superscript a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis. Standard errors are clustered by zipcode and original vintage year.

additional check, I re-estimate the full-control specification after replacing origination vintage fixed effects with reporting-time fixed effects at the month, quarter, or year level. These fixed effects absorb aggregate refinancing cycles, current interest-rate environments, and other macroeconomic conditions common to all borrowers observed at the same point in time. The income gradient remains positive, monotone, and highly statistically significant under both the simple and ADL-adjusted in-the-money definitions. In the most stringent monthly fixed-effect specification, the top-quintile coefficient is 1.48 percentage points under the simple rate-gap definition and 2.02 percentage points under the ADL-adjusted definition (see Table A.12). Thus, the results indicate the baseline pattern is not driven by time-varying aggregate refinancing conditions.

3.2 Refinancing Probability by Rate Gap

The average refinancing probabilities in Section 3.1 condition on in-the-money status but pool across the full distribution of rate gaps. Disaggregating by rate gap adds two layers of insight.

First, it sharpens the test of behavioral heterogeneity: by holding the size of the financial incentive constant within each bin, the comparison across income groups isolates differences in the response to a given gain rather than any level differences in the rate gaps income groups happen to face.

Second, the shape of the income gradient across rate-gap bins is diagnostically informative about the underlying mechanism. If lower-income households underperform only at marginal rate gaps — where uncertainty about net refinancing benefits might rationalize inaction — the pattern would be consistent with rational caution. If instead the income gradient widens precisely where the financial case for refinancing is strongest, at large positive gaps, the evidence points toward inattention or execution frictions that prevent households from acting on clear-cut opportunities. The out-of-the-money region provides a natural benchmark and implicit falsification: income groups that face no rate benefit from refinancing should behave similarly regardless of income, and departures from this baseline should emerge only once a financial gain materializes.

To quantify how refinancing behavior varies across income groups conditional on refinancing incentives, I estimate the following specification:

$$Refi\_Indicator_{it} = \alpha + \sum_{Quintile \in Q} \sum_{Bin \in K} \beta_q^k \cdot \mathbf{1}[RateGap_{it} \in k] \cdot \mathbf{1}[IncomeQuintile_i \in Q] \tag{3}$$

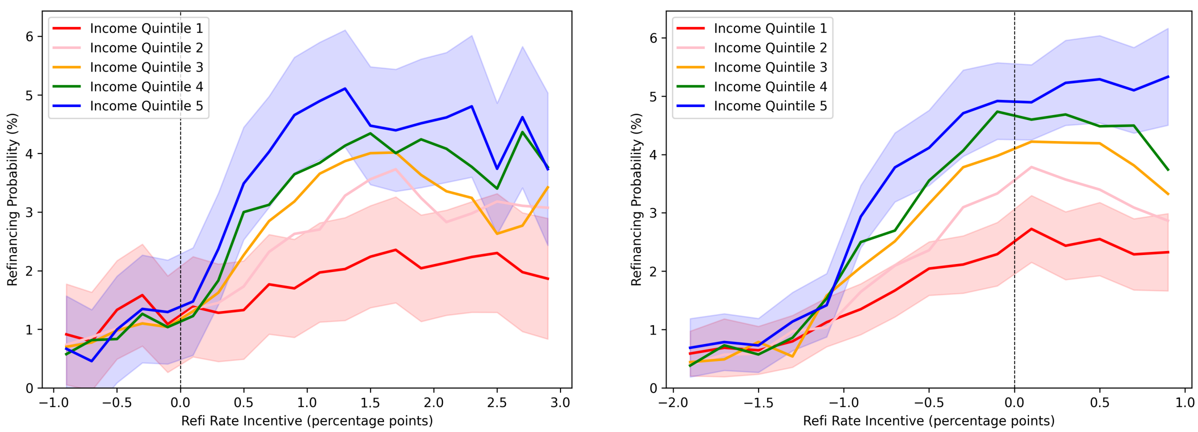
$$+ \Gamma' X_{it} + \nu_{z(i)} + \lambda_{v(i)} + \epsilon_{it} \tag{4}$$

I controlled the same variables as the previous regression using a non-parametric regression as specified in Equation 3. Again, $Refi\_Indicator_{it}$ is an indicator variable equal to 1 when the loan i is refinanced in month t. k denotes rate-gap bins and j is the five income quintile at the time of origination. $\mathbf{1}[RateGap_{it} \in k]$ is a dummy variable equal to 1 when the rate gap falls in bin k, and X_{it} is a series of control variables. $\nu_{z(i)}$ and $\lambda_{v(i)}$ are ZIP code and vintage year-quarter fixed effects, respectively, and standard errors clustered by zip code and original vintage year.

The left panel of Figure 3 presents the refinancing probability estimated using the simplest identification rule, where the threshold is set to zero. As shown, in the out-of-the-money region, refinancing rates are broadly similar across income quintiles. However, once loans enter the in-the-money region, a clear heterogeneity emerges. Even after accounting for the confidence bands of the first and fifth quintiles, high-income households are significantly more likely to refinance than low-income ones.

The right panel applies a more robust identification method based on the *ADL-adjusted rate gap*, which allows the refinancing threshold to vary by loan and time (i, t). It is worth noting that this plot does not simply represent a horizontal shift of the left panel; rather, the bin assignment changes dynamically for each observation as the threshold evolves. This conservative specification already reveals refinancing inequality beginning around minus one percentage point, and the gap widens further in the in-the-money region.

Overall, regardless of the specific identification method, the pattern of refinancing inequality is consistently observed across nearly all rate-gap bins.



(a) Simple Rate Gap ($\theta_{it} = 0$) (b) ADL-adjusted Rate Gap ($\theta_{it} = ADLgap_{it}$)

Figure 3: Refinancing Inequality by Rate Gap

3.3 Cumulative Refinancing within Twelve Months

The monthly hazard estimates above measure the flow probability of refinancing in a given month conditional on being in-the-money. While informative, this flow measure can mask economically large differences in cumulative take-up: small differences in monthly hazards compound over time, and the relevant policy and welfare object is often how many borrowers act on a refinancing opportunity within a reasonable horizon rather than the per-month rate at which they do so. To translate the hazard estimates into an interpretable take-up measure, I next examine the cumulative probability that a loan refinances within twelve months of first becoming in-the-money.

For each loan, let T_i^{ITM} denote the first month in which loan i becomes in-the-money. I define

$$Refi\_Indicator\_12m_i = 100 \times \mathbf{1}\left\{\sum_{\tau=0}^{11} Refi\_Indicator_{i,T_i^{ITM}+\tau} > 0\right\}.$$

I estimate a linear probability model regressing $Refi\_Indicator\_12m_i$ on income-quintile indicators, with the bottom quintile omitted, and the same borrower and loan controls, ZIP-code fixed effects, and origination vintage year-quarter fixed effects used in the baseline monthly specification. Unlike the monthly LPM, this is an event-time regression: each loan contributes a single observation indexed by its first in-the-money entry, and the dependent variable summarizes its refinancing behavior over the subsequent twelve months. I report results under both the simple rate-gap and ADL-adjusted rate-gap definitions of being in-the-money.

Table 3: Twelve-Month Cumulative Refinancing Probability by Income Quintile

Dependent	Refinancing within 12 Months × 100	
ITM Definition	Simple Rate Gap	ADL Gap
Income Quintile 1	17.24[a]	23.59[a]
	(0.38)	(0.62)
Income Quintile 2	18.54[a]	28.58[a]
	(0.33)	(0.48)
Income Quintile 3	19.81[a]	31.11[a]
	(0.33)	(0.30)
Income Quintile 4	21.49[a]	32.03[a]
	(0.43)	(0.52)
Income Quintile 5	24.80[a]	34.91[a]
	(0.63)	(0.69)
Q5–Q1 gap	7.57	11.32
Q5/Q1 ratio	1.44	1.48
Observations	326,147	81,467

Notes: Superscript a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis. This table reports predicted probabilities from linear probability models of refinancing within twelve months after a loan first becomes in-the-money. Each loan contributes one observation indexed by its first in-the-money entry. The dependent variable is scaled by 100, so entries are in percentage points. The full-control specification includes borrower and loan controls, ZIP-code fixed effects, and origination vintage year-quarter fixed effects. Standard errors are clustered by ZIP code and original vintage year.

Table 3 reports the main estimates. Under the ADL-gap definition, the predicted twelve-month refinancing probability rises monotonically from 23.6 percent in the bottom income quintile to 34.9 percent in the top quintile: a top-to-bottom ratio of approximately 1.48. The coefficient on each higher income quintile is positive, monotone, and statistically significant at 1% levels. Results under the simple rate-gap definition show qualitatively similar top-to-bottom ratio: the predicted twelve-month probability rises from 17.2 percent in the bottom quintile to 24.8 percent in the top quintile, a ratio of 1.44. In this specification, all the upper three quintiles are strongly statistically different from the bottom quintile. Full regression estimates are reported in Appendix Table A.13.

These estimates have an economically intuitive interpretation. Within one full year after a refinancing opportunity first arises, lower-income borrowers are substantially less likely to have acted on it. Under the ADL-gap definition, roughly three quarters of bottom-quintile borrowers remain in their original mortgage twelve months after first becoming in-the-money, compared with about two thirds of top-quintile borrowers. This pattern indicates that the income gradient in monthly hazards documented above is not merely a statistical artifact of small per-month differences: it compounds into a large cross-income inequality in the cumulative take-up of refinancing opportunities.

The twelve-month window provides an economically interpretable summary of take-up, but does not characterize the full distribution of refinancing timing. The next subsection addresses this distinction directly using event-time survival curves.

3.4 Delays in Refinancing Decisions

The foregoing analysis documents that lower-income borrowers are less likely to refinance when it is financially beneficial to do so — both in their monthly hazard and in cumulative take-up within twelve months of first becoming in-the-money. The existing literature on heterogeneous refinancing behavior has focused almost exclusively on this *intensity* margin: how much borrowers refinance when refinancing is profitable. Yet because refinancing is an option-like decision, when borrowers choose to exercise the option is a distinct and equally consequential dimension of heterogeneity — one that has received comparatively little systematic attention. This paper provides, to my knowledge, the first loan-level evidence that heterogeneous refinancing in *timing*: lower-income borrowers are not only less likely to refinance but also systematically slower to act when they do. The timing margin carries independent welfare significance: even borrowers who eventually refinance forgo months of payment savings if they act late, a cost that intensity measures alone cannot capture.

To examine this, I align all loans at the month in which they first become in-the-money and compute, for each income quintile, the share of loans that remain outstanding over time. This approach yields a monthly survival rate of outstanding mortgages relative to the in-the-money point.[8] In this analysis, I use a more robust measure of being in-the-money based on the ADL-adjusted rate gap, as described in Section 2.4. This measure is particularly important for analyzing refinancing timing, because the financial gain from refinancing declines sharply as a mortgage approaches maturity. The ADL-based measure incorporates this feature and therefore provides a more accurate basis for evaluating refinancing behavior over time.

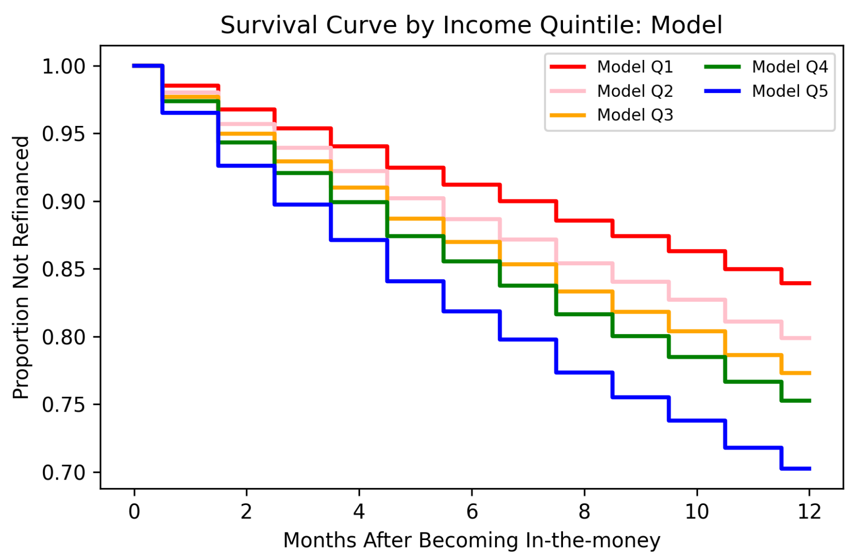


Figure 4: Proportion of Remaining Loans by Months After Becoming In-the-money

Figure 4 presents the resulting survival curves. Starting about two months after entering the money, loans held by households in the top income quintiles exhibit noticeably lower survival rates—meaning they refinance more quickly—than those in the bottom quintiles. This timing

[8]To implement this approach cleanly, I account for an important data feature. A complication arises because a loan may fall out-of-the-money shortly after first becoming in-the-money. In such cases, I treat these in-the-money occurrences as part of the same episode as long as the time gap between consecutive in-the-money months is less than twelve months. For example, suppose a loan becomes in-the-money and begins an episode, remains in that state for several months, then briefly falls out of the money, and becomes in-the-money again six months later. I treat this entire sequence as a single in-the-money episode, since the interval between the two in-the-money observations is less than twelve months.

gap is persistent and economically meaningful. For example, the top income quintile reaches a 20 percent refinancing rate – corresponding to 80 percent of loans remaining outstanding – after roughly four months, whereas the bottom quintile requires approximately ten months to reach the same threshold. The survival patterns are also quantitatively consistent with the refinancing probabilities reported in Table 3. After twelve months in the in-the-money state, the estimated survival rate is 0.756 for the bottom income quintile and 0.564 for the top quintile, implying cumulative refinancing rates of approximately 24.4 percent and 43.6 percent, respectively. These magnitudes closely mirror the refinancing gradients estimated in the baseline panel regressions, reinforcing the robustness of the estimated income gradient across empirical approaches.

3.5 Robustness Checks

Lender-side constraints and origination-lender sorting. Concerns may remain that the observed refinancing inequality is driven primarily by lender-side underwriting constraints rather than borrower-side refinancing behavior. While the baseline specification already controls for a rich set of borrower and loan characteristics, I further assess this channel by re-estimating the baseline regression within credit-score quartiles, a key underwriting input. The results show that the income gradient in refinancing remains pronounced within each credit-score bin: higher-income borrowers refinance at substantially higher rates than lower-income borrowers, even among borrowers with similar credit scores (see Figure C.4).

I also examine whether affordability constraints can explain the gradient by splitting the sample into payment-to-income (PTI) bins corresponding to common underwriting thresholds. Re-estimating the baseline specification within each PTI bin yields a similar pattern: the estimated income-quintile differences are broadly stable across PTI ranges, and remain statistically meaningful except in bins with limited observations. Importantly, the income gradient persists even in the most underwriting-favorable group with PTI below 36 percent, where approval constraints should be least binding (see Table C.14).

Finally, I use the Freddie Mac-HMDA linked panel to identify the origination lender of each matched loan. This allows me to address the concern that low-income borrowers were originated by different lenders, or by different lender-year vintages, and that those lender or vintage differences explain later refinancing inequality. The HMDA-linked sample identifies origination lenders for 90,637 loans from 6,828 lenders over the 2000–2022 origination cohorts; the in-the-money regression samples contain about 1.78 million monthly observations under the simple rate-gap definition and about 342,000 monthly observations under the ADL-gap definition. Re-estimating the full-control specification with origination lender-year fixed effects leaves the income gradient positive and highly significant for both the simple rate-gap and ADL-gap samples (see Appendix C.2). The gradient remains within the same origination lender-year cells, suggesting that the result is not driven by initial lender composition, origination pricing environments, or lender-vintage-specific underwriting conditions. These specifications do not directly observe refinance-stage outreach or screening, but they substantially weaken the explanation that the benchmark estimates are mechanically produced by sorting into original lenders.

Changes in income quintile. Another major concerns is that since income quintiles are observed at the time of origination, changes in income quintiles at the time of refinancing might cause bias in regression results. Given that income quintile is a key variable, I performed four different robustness checks to validate the previous results.

First, I calculated the average loan age up to the refinancing point and used the Panel Study of Income Dynamics (PSID) data to track changes in mortgagors' income quintiles over the same period. In my sample, the average time to refinancing was 23 months. Therefore, I examined the changes in mortgagors' income quintiles over two years using the same yearly-changing income quintile thresholds. The probability of the lowest income quintile rising to the highest quintile after two years was 5.6%, while the probability of the highest quintile falling to the lowest quintile was 2.4%, both very low as suggested in Table C.18.[9] Although the data's limitations prevent tracking income changes, these results suggest that my estimates are unlikely to be significantly distorted by frequent changes in income quintiles.

[9] Broadening the scope, the probability of the bottom two (1st and 2nd) income quintiles remaining in the same two quintiles after two years was 72.7%, whereas the probability of changing to the top two (4th and 5th) quintiles was 12.6%. Conversely, the probability of the top two income quintiles remaining in the same quintiles was 81.6%, while the probability of falling to the bottom two quintiles was 8.2%, indicating a low likelihood.

Second, my data panel matches old loans with new loans, allowing for the observation of income and other characteristics at the time of refinancing. Therefore, I was able to perform the same regression using Equation 2 with income quintile and other control variables at the time of refinancing. The results confirmed that the pattern consistent with the benchmark regression was maintained. As shown in Table C.19, even with a full set of control variables, the predicted refinancing probability for high-income households remained significantly higher than for lower income quintile groups.

A caveat is that this refi-time-income specification is estimated only on matched refinance pairs, unlike the benchmark loan-month panel that also includes loans that never refinance. As a result, the predicted monthly refinancing probabilities are mechanically higher than in the benchmark specification. I also do not include the HARP indicator in this exercise: when HARP status is defined from the new refinance loan, HARP refinances are concentrated almost entirely in the bottom two income quintiles, so absorbing HARP would mechanically remove much of the low-income refinancing variation. I therefore interpret this exercise as a check on income-quintile reclassification, rather than as a separate HARP-control specification.

Third, I conducted a regression analysis on only those loans where there was no change in income quintile between the new and old loans. The results still showed a clear refinancing inequality, as illustrated in Table C.20. This indicates that the previous results hold even when excluding changes in income quintile.

Lastly, since the probability of changes in a borrower's income quintile is lower for loans with shorter ages, I conducted repeated regression analyses by varying the age of the sample loans. The results showed a clear refinancing inequality regardless of the loan age cut. Table C.20 illustrates the predicted probabilities from regression (5), which includes all control variables, analyzed by progressively widening the loan age range: less than one, two, three, five, and seven years. As the loan age sample increased from one year or less to two years or less, there was an overall increase in probability, causing a line shift. However, it was still evident that higher income was significantly associated with higher refinancing probabilities. The same pattern persisted when increasing the loan age limit further. Additional robustness checks using the Survey of Consumer Finances (SCF) are reported in Appendix C.4.

3.6 Unrealized Savings from Refinancing

Having identified significant refinancing inequality across income groups, I now investigate the extent to which this results in missed increases in disposable income. This study aims to examine how refinancing inequality ultimately impacts the monetary policy transmission channel, making the quantification of the saving amount crucial. I conducted the analysis on how much households' disposable income could have increased conditional on refinancing.

To start this analysis, I switched to a matched pair dataset of old and new loans. I analyzed how much each income group could reduce their interest rates through refinancing using a linear probability model. Using Equation 2, I changed the dependent variable as the difference between the interest rate of the new loan and that of the old loan, leading to the specification of Equation 5. I restrict the sample to in-the-money observations, since the calculation of missing gains is only meaningful when refinancing would have generated a financial benefit.

$$OldRate_i - NewRate_i = \alpha + \sum_{q=2}^{5} \beta_q \cdot IncomeQuintile_{ij} + \Gamma' \cdot X_{it} + \nu_{z(i)} + \lambda_{v(i)} + \epsilon_{it} \qquad (5)$$

Table 4: Unrealized Savings from Refinancing by Income Quintile

Income Quintile	Avg. Income ($)	Avg. Balance ($)	Est. Rate Gap (pp)	Mthly. Pymt. Reduction ($)	Reduction / Income (%)
1	16,707	91,842	1.07	104.18	7.61
2	28,175	147,917	1.06	164.73	6.79
3	39,441	197,424	1.05	217.76	6.33
4	54,884	256,552	1.05	283.44	5.76
5	97,976	316,852	1.07	409.54	4.37

Notes: Payment savings assume refinancing resets the loan to a 30-year FRM term. Rate gaps are estimated using Eq. (5) (full controls). Freddie Mac statistics suggest that about 70–75% of refinances keep the term length unchanged. Sample is restricted to in-the-money observations ($gap > 0$).

Based on the estimated rate reductions, I performed a back-of-the-envelope calculation to estimate the potential savings. By subtracting the rate reduction attributed solely to income quintiles from the existing interest rates, I derived the new post-refinancing interest rates. Assuming all refinancing are done with a 30-year fixed-rate mortgage, I calculated the monthly payment reduction using these new estimated rates and examined this as a proportion of income.

The results Table 4 shows that the bottom two income quintiles could have saved 6.8-7.6% of their monthly income through refinancing. This is a significant amount, as it represents additional disposable income occurring every month, not just a one-time saving. Although the absolute dollar amount of missed gains can be larger for higher-income borrowers—primarily because they tend to hold larger mortgage balances—the foregone savings constitute a much larger share of income for lower-income mortgagors. Given their higher marginal propensity to consume, the missing gains for these households likely translate into substantial unrealized consumption responses.

As a robustness check, I also estimate unrealized savings using an alternative approach: for each in-the-money loan-month, I construct a counterfactual refinancing rate by subtracting the income-group-specific predicted rate gap—estimated from Equation 22—from the borrower's current coupon, and translate the implied payment reduction into dollars using loan-specific balance and maturity. This alternative procedure yields similar estimates, with the bottom quintile facing foregone monthly savings of approximately 7.6% of income, corroborating the findings from the matched-loan approach. Together, these findings suggest that refinancing inequality could materially weaken the pass-through of monetary policy to disposable income and consumption, precisely among households with the highest marginal propensity to consume. (see Appendix C.5 for details).

As a complementary check, **??** examines refinancing responses directly around high-frequency monetary policy shocks using a regional panel constructed from the Freddie Mac data. This exercise is not part of the main identification strategy, since monetary policy shocks are common aggregate shocks and the effective time-series dimension is limited, making inference sensitive to time clustering. Nevertheless, the point estimates display the same monotone income gradient as the loan-level evidence: refinancing responses to a 1 percentage point rate cut rise from the bottom to the top income quintile. I therefore interpret the regional local-projection results as suggestive macro-level validation that the refinancing inequality documented in the loan-level panel also appears in the response to monetary policy shocks.

3.7 Differential Refinancing Response to Monetary Policy Shocks

The evidence in the previous sections document refinancing inequality using a loan-level panel that conditions on each borrower's in-the-money status. This cross-sectional approach isolates the friction channel cleanly, but it does not speak directly to a distinct question: do lower-income borrowers also respond less to the same monetary policy shock in real time, when the Federal Reserve cuts rates and refinancing opportunities open up for the broad market? A large literature documents the *aggregate* dynamic refinancing response to monetary policy (Eichenbaum et al. 2022; Beraja et al. 2019; Wong 2021), and a separate literature documents *cross-sectional* heterogeneity in refinancing rates (Andersen et al. 2020; Gerardi et al. 2023; Agarwal et al. 2024).

To my knowledge, however, prior work has not directly estimated how the *dynamic refinancing response* to identified monetary-policy shocks varies across borrowers. This distinction is important: documenting heterogeneous refinancing propensities is not the same as tracing heterogeneous impulse responses to exogenous monetary-policy surprises. By combining plausibly exogenous monetary-policy shocks, dynamic impulse-response estimation, and borrower-group heterogeneity, this section connects refinancing inequality more directly to the transmission of monetary policy at the macro level.

I construct a quarterly regional panel from the Freddie Mac loan-level data following the aggregation approach in Eichenbaum et al. (2022). For each three-digit ZIP-code region, quarter, and income quintile, I compute the stock of outstanding fixed-rate, single-family, owner-occupied mortgages and the number of rate refinances originated in that quarter. The outstanding-loan stock is constructed from the performance panel, while refinancing originations are counted from the origination data. I exclude HARP refinances from the numerator and begin the estimation sample in 2004 to allow the outstanding-loan stock to build up after the start of the Freddie Mac sample. The monetary-policy shocks are high-frequency surprises from Swanson (2021), which decompose

FOMC announcement-window asset-price movements into federal funds rate, forward-guidance, and LSAP components.

The dependent variable is a cumulative refinancing response. Let x_{zqt} denote the number of refinances in ZIP3 region z, income quintile q, and quarter t, and let y_{zqt} denote the corresponding outstanding mortgage stock. My preferred measure is the cumulative sum of quarterly refinancing rates,

$$\rho^h_{zqt} = \sum_{k=0}^{h} 100 \times \frac{x_{zq,t+k}}{y_{zq,t+k}}, \tag{6}$$

which is directly comparable to the refinancing-rate construction in Eichenbaum et al. (2022). This measure asks how many mortgages refinance, as a share of the outstanding stock in each quarter, over the h-quarter window following a monetary-policy shock.

To estimate the income gradient, I stack the five income-quintile cells within each ZIP3-quarter and estimate an interaction specification that absorbs both ZIP3-by-income fixed effects and quarter fixed effects. For each horizon h, I estimate

$$\rho^h_{zqt} = \alpha^h_{zq} + \lambda^h_t + \beta_h \left(\Delta m_t \times \tilde{q}\right) + \sum_{\ell=1}^{6} \Gamma^h_\ell W_{zq,t-\ell} + \sum_{\ell=0}^{6} \Pi^h_\ell Z_{z,t-\ell} + \varepsilon^h_{zqt}. \tag{7}$$

Here Δm_t is the change in the 30-year mortgage rate, normalized so that a positive value corresponds to a decline in rates. The variable $\tilde{q}$ is a normalized income rank, defined as $\tilde{q} = (q-1)/4$, so that $\tilde{q} = 0$ for the bottom income quintile and $\tilde{q} = 1$ for the top quintile. The coefficient β_h is therefore directly interpretable as the Q5–Q1 differential refinancing response to a 1 percentage point monetary easing. The specification includes ZIP3-by-income-quintile fixed effects α^h_{zq} and quarter fixed effects λ^h_t. Quarter fixed effects absorb all aggregate dynamics, including the average effect of monetary policy, business-cycle conditions, and aggregate mortgage-market conditions. Identification therefore comes from whether higher-income cells within the same quarter respond more strongly to the monetary shock than lower-income cells.

I instrument $\Delta m_t \times \tilde{q}$ and its lags using the monetary policy surprises interacted with $\tilde{q}$. The controls $W_{zq,t-\ell}$ include lagged refinancing rates, and $Z_{z,t-\ell}$ includes local house-price growth, local unemployment, aggregate output growth, and SPF-based macroeconomic expectations to control for the effect of the "information channel", following Nakamura and Steinsson (2018). All specifications include six quarterly lags. Regressions are weighted by the outstanding loan stock in each ZIP3-income cell, and standard errors are two-way clustered by ZIP3 region and quarter.

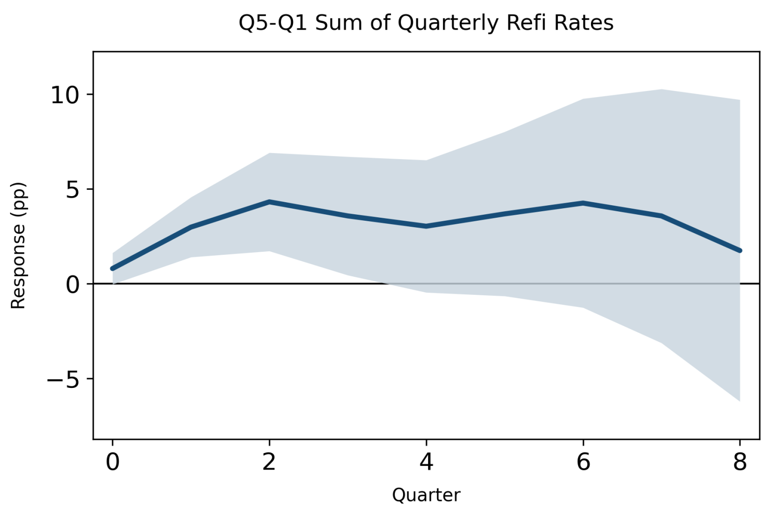


Figure 5: Q5–Q1 Differential Refinancing Response to a Monetary Policy Easing

Notes: The figure plots the coefficient β_h from Equation 7. The dependent variable is the cumulative sum of quarterly refinancing rates defined in Equation 6. The monetary-policy shock is normalized so that a positive value corresponds to a 1 percentage point decline in the 30-year mortgage rate. The shaded region denotes 90 percent confidence intervals based on two-way clustering by ZIP3 region and quarter.

Figure 5 reports the estimated Q5–Q1 differential response. Two features stand out. First, the economic magnitude of the differential response is substantial: the gap peaks at approximately 4.3 percentage points after two quarters, which is large relative to the average aggregate refinancing response of about 2.7 percentage points at the same horizon. This comparison suggests that the aggregate refinancing response to monetary easing is disproportionately concentrated among higher-income borrowers. Second, the income gradient is similar across two distinct empirical strategies: the borrower-level rate-gap design in the previous section and the monetary-shock interaction design in this section. This consistency strengthens the interpretation that refinancing inequality reflects behavioral and financial frictions, rather than confounders specific to either empirical approach. Third, this pattern – a large, immediate cross-income differential that decays within roughly four quarters – is consistent with the model's heterogeneous refinancing hazards Figure 7 and provides a reduced-form counterpart to the structural decomposition in Section 4.

Appendix D reports the first-stage estimates, the average refinancing response, and robustness checks showing that the income gradient is stable across alternative refinancing-response measures and income-group definitions.

4 A Model for Heterogeneous Refinancing Friction

To quantitatively assess how the empirical refinancing gaps documented above affect the transmission of monetary policy, this paper develops a structural model of household mortgage refinancing. The model builds on the framework of Berger et al. (2021) , but departs from the baseline specification in an important dimension. In the original model, refinancing inequality does not emerge: instead, lower-income households—due to their limited liquid asset holdings—behave as wealthy hand-to-mouth agents and participate more actively in refinancing, generating a pattern opposite to that observed in the data. By incorporating three additional mechanisms described below, the proposed model successfully reproduces the empirically documented refinancing inequality, whereby higher-income households refinance more frequently while lower-income households exhibit systematically lower refinancing intensity.

The model improves upon the existing framework along three key dimensions. First, while the baseline model relies solely on an inattention channel to rationalize low refinancing rates, this paper distinguishes between two conceptually different sources of refinancing frictions: inattention and hassle costs. Separating these frictions allows the model to capture distinct behavioral and economic mechanisms underlying refinancing decisions. Second, the baseline specification models attention as a simple on–off switch depending on whether a household is in-the-money, which fails to capture how attention varies with the magnitude of the rate gap. This limitation is quantitatively important, as the strength of monetary policy transmission may depend on the size of interest rate cuts. To address this, the model introduces a gap-dependent attention intensity rather than a binary attention mechanism. Third, and most importantly, the model allows both inattention and hassle cost frictions to vary by income state. Attention heterogeneity is disciplined using a parsimonious parametric structure consistent with empirical patterns observed in the data, while refinancing thresholds associated with hassle costs are estimated fully nonparametrically.

4.1 Model setup

The economy is populated by a continuum of risk-averse households who solve an infinite-horizon optimization problem in continuous time. Households face both aggregate interest rate risk and idiosyncratic labor income shocks. The aggregate short rate follows an exogenous process and is controlled by the monetary authority. Households are heterogeneous with respect to labor assets, outstanding mortgage balances, coupon rates, and income. Each household holds a long-term fixed-rate mortgage and can choose whether to refinance, including the option to extract home equity through cash-out refinancing. Refinancing requires paying a fixed cost and, in addition, households face two state-dependent refinancing frictions that are described below. On the supply side, lenders are risk-neutral. Taking the exogenous short rate as given, lenders price mortgage contracts based on expected prepayment behavior summarized by endogenous hazard rates, and supply mortgages such that equilibrium mortgage rates clear the mortgage market

4.2 Mortgage Refinancing and Frictions

Mortgage refinancing in the model is governed by two distinct and complementary frictions, which I decompose to isolate their separate economic roles. The first is a *threshold-type friction*, capturing the fact that refinancing entails not only explicit monetary fixed costs but also non-monetary costs, such as time costs, information-processing costs, and search costs for better mortgage terms. If this friction were the only impediment to refinancing, households would refinance with probability one whenever the gains from refinancing exceed a sufficiently high threshold.

The second friction is *inattention*. Even when refinancing gains are large enough to overcome threshold costs, observed refinancing rates remain far below one. This pattern suggests that households do not continuously monitor refinancing opportunities, but instead pay attention only intermittently. The model captures this behavior through a Calvo-style attention friction.

By decomposing refinancing frictions into an execution margin (threshold) and an attention margin (frequency), the model allows us to separately quantify the contribution of each channel to refinancing activity and to conduct counterfactual exercises that shut down individual frictions. This decomposition is central to designing policies aimed at reducing refinancing frictions and understanding how such frictions shape the transmission of monetary policy.

Hassle-cost channel. The execution channel is modeled as an income-dependent refinancing threshold. A household executes a refinancing transaction only if the value gain from refinancing exceeds an income-specific threshold. To avoid imposing restrictive functional form assumptions, these thresholds are estimated fully nonparametrically across income groups. The resulting execution rule is summarized in equation (8), which characterizes refinancing as a comparison between the value gain from refinancing and the income-dependent hassle-cost threshold.

$$\rho^{refi} = \begin{cases} 1, & \text{if } V^{\text{refi}} - V^{\text{stay}} > \Phi_s \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

Attention friction. The attention channel governs how frequently households pay attention to refinancing opportunities over calendar time. Existing models typically assume that attention switches discretely on or off depending on whether a household is in-the-money. However, this binary formulation fails to capture a key behavioral feature of refinancing behavior: households naturally become more attentive to refinancing as the gap between their current mortgage coupon rate and the prevailing market rate widens. This distinction is quantitatively important, as the strength of monetary policy transmission may depend on the magnitude of interest rate movements rather than solely on whether refinancing is profitable.

In the model, attention varies continuously with the rate gap. Households receive opportunities to consider refinancing according to a Poisson process, whose arrival intensity depends on both the rate gap and the household's income state. Attention intensity increases as the household becomes deeper in-the-money.

The functional form of attention is disciplined by two robust empirical patterns observed in the data. First, even for households that are deep in-the-money, attention to refinancing remains limited rather than approaching one. Second, the maximum level of attention differs systematically across income groups. These features are captured by a parsimonious, gap-dependent attention specification, formalized in equation (9), which allows for income-dependent heterogeneity in maximum attention capacity.

$$\lambda^A(g_t, s_t) = \chi_{\max}(s_t) \frac{1}{1 + \exp\left(-k\,[\,g_t - \mu\,]\right)} \tag{9}$$

Here, $g_t \equiv c_t - m(r_t)$ denotes the difference between the household's current mortgage coupon rate (c_t) and the prevailing market mortgage rate $m(r_t)$. $\lambda^A(g_t, s_t)$ is the Poisson arrival intensity for attention to refinancing opportunities. $\chi_{\max}(s_t)$ denotes the maximum attention capacity for households in income state s, capturing the upper bound on how frequently a household can pay attention to refinancing opportunities even when it is deep in-the-money. The parameter μ governs the location of the attention response, representing a region in which rate gaps are too small to meaningfully trigger attention. The slope parameter k controls how sensitively attention responds to changes in the rate gap, that is, how rapidly attention intensity increases as the gap widens.

4.3 Household Problem

Households maximize lifetime utility by choosing both their consumption (C_t) path, the mortgage refinancing (ρ_t^{refi}), and cash-out decision (ΔF_t^{co}). These decisions are made subject to the structure of the mortgage contract and the household's intertemporal budget constraint. Given an income process and the prevailing interest rate environment, each household dynamically decides how much to consume today and when to refinance its existing mortgage.

$$V(S_t, r_t) = \sup_{\{C_t, \rho_t^{refi}, \Delta F^{co}\}} \mathbb{E}_t \left[\int_0^\infty e^{-\delta t} \frac{C_t^{1-\gamma}}{1-\gamma} \, dt \right],$$

$$\text{s.t.} \quad dW_t = (Y_t - C_t + r_t W_t - c_t F_t)\, dt + \underbrace{dN_t^A \rho_t^{refi}}_{\text{refi decision}} \big(\underbrace{\Delta F_t^{co}}_{\text{cash-out}} - \underbrace{\kappa}_{\text{refi fixed cost}} \big),$$

$$dF_t = \underbrace{-\alpha F_t \, dt}_{\text{amortization}} + \underbrace{(F_t' - F_t) dN_t^A \rho_t^{refi}}_{\text{refi jump}},$$

$$dc_t = \underbrace{(m(r_t) - c_t)\big(dN_t^m + dN_t^A \rho_t^{refi}\big)}_{\text{coupon reset at refinancing (moving + refi)}} .$$

The value function $V(S_t, r_t)$ represents a household's maximal expected lifetime utility given its current state. The individual state is $S_t \equiv (W_t, F_t, c_t, s_t)$, where W_t denotes liquid wealth, F_t the outstanding mortgage balance, c_t the existing mortgage coupon rate, and s_t the idiosyncratic income state. The aggregate state of the economy is summarized by the short rate r_t. The vector $\boldsymbol{\vartheta}$ collects the calibrated model parameters. The attention channel is governed by $\chi_{\max}(s)$, k, and μ, while the execution channel is governed separately by the income-specific adjustment thresholds.

Households choose consumption C_t and refinancing-related actions, including whether to execute refinancing and the size of cash-out refinancing ΔF_t^{co}, to maximize discounted expected utility over an infinite horizon. Preferences are represented by a CRRA utility function with risk aversion parameter γ, and future utility is discounted at rate δ.

Refinancing opportunities arrive stochastically. In particular, households receive opportunities to consider refinancing according to a Poisson process dN_t^A with intensity $\lambda^A(g_t, s_t)$, where the intensity depends on the rate gap and the current income state. Conditional on such an arrival, households decide whether to execute refinancing subject to the execution (hassle-cost) threshold. In addition, exogenous events such as relocation or other forced mortgage resets trigger refinancing according to an independent Poisson process dN_t^m. These arrival processes jointly govern the timing of mortgage refinancing and coupon resets in the model.

It is important to emphasize that income affects refinancing behavior in a *non-monotonic* manner, even absent any mechanical targeting of refinancing frictions. On the one hand, lower-income households are more likely to be constrained borrowers. They face tighter payment-to-income (PTI) constraints and may lack the liquidity required to cover fixed refinancing costs. As a result, even when refinancing opportunities arise, these constraints limit their ability to act, giving rise to a *constraint channel* that lowers refinancing probabilities among low-income households.

On the other hand, lower-income households exhibit a strong *liquidity motive*. Because they place a higher value on immediate liquidity, they have stronger incentives to extract home equity through cash-out refinancing. Conditional on being able to refinance, they therefore tend to refinance more aggressively. The overall effect of income on refinancing and cash-out policy functions is thus non-monotonic, reflecting the interaction between these two opposing forces rather than a mechanical consequence of the model's frictional structure.

4.4 Uncertainty Environment

The model features two sources of uncertainty. First, households face idiosyncratic labor income risk, following the AR(1) process:

$$d \ln Y_t = \eta_y \, dt + \sigma_y \, dZ_t^y \tag{10}$$

Each household's labor income Y_i evolves stochastically according to a mean-reverting process with persistence parameter η_y and volatility σ_y. The innovation is driven by a standard Brownian motion Z_t^r that is independent across households and orthogonal to all aggregate shocks. This idiosyncratic income component introduces cross-sectional heterogeneity even in the absence of aggregate fluctuations.

Second, the aggregate short-term interest rate r_t follows a continuous-time stochastic process of the Cox–Ingersoll–Ross (CIR):

$$dr_t = -\eta_r (r_t - \bar{r}) \, dt + \sigma_r \sqrt{r_t} \, dZ_t^r, \tag{11}$$

It is characterized by persistence η_r, volatility σ_r, and long-run mean $\bar{r}$. The process is driven by a common Brownian motion Z_t^r, which captures aggregate shocks affecting all households simultaneously. Changes in r_t propagate through to mortgage rates and rate gaps, thereby linking aggregate monetary conditions to household-level refinancing dynamics.

Together, the idiosyncratic income process (Y_i) and the aggregate short-rate process (r_t) generate both individual and aggregate uncertainty in the model. These shocks jointly shape the evolution of consumption, wealth accumulation, and mortgage refinancing behavior across the household distribution.

4.5 Mortgage Market Equilibrium

Each mortgage is modeled as a fixed-rate loan contract characterized by its coupon rate c and outstanding balance F. Instead of specifying a detailed amortization schedule, the model assumes that a constant fraction α of the loan is repaid each year. This simplification implies that repayments accelerate slightly as the loan approaches maturity, but it greatly reduces computational complexity without affecting the key results. When a household refinances, the coupon is reset to the prevailing market mortgage rate $m(r_t)$.

The mortgage price satisfies a risk-neutral pricing condition that accounts for the expected prepayment probability. Lenders fund themselves at the short rate and earn expected returns from mortgage payments such that, in equilibrium, they break even. In other words, the present value of a newly issued mortgage equals its face value, and the equilibrium coupon $m(r_t)$.

A key assumption is that lenders do not observe each household's refinancing friction. Instead, they form expectations based on the average refinancing hazard across the population ($\bar{\boldsymbol{\vartheta}}$).[10]

$$P(W, F, r, c, s; \bar{\boldsymbol{\vartheta}}) = \mathbb{E}\left[\int_0^{\tau} e^{-\int_0^t r_d\,du}\, c\,dt + e^{-\int_0^{\tau} r_d\,du} \,\middle|\, r_0 = r \right] \tag{12}$$

[10]As noted by Berger et al. (2024), households with higher attention parameters are overrepresented among those who actually prepay, relative to the ex-ante distribution. This discrepancy between ex-ante and ex-post distributions can affect the lender's perceived hazard rate and, consequently, mortgage pricing. In the current version of the model, this feedback is not yet incorporated but will be addressed in a later refinement of the framework.

Under the assumption of complete markets, the equilibrium mortgage rate is determined such that the lender's expected profit is zero after accounting for anticipated refinancing behavior.

$$P(W, F, r, c, s; \bar{\boldsymbol{\vartheta}}) = 1 \tag{13}$$

4.6 Dynamic Programming and Distributional Dynamics

Hamilton–Jacobi–Bellman Equation. Given the continuous-time formulation and the presence of stochastic refinancing opportunities, the household's optimization problem admits a Hamilton–Jacobi–Bellman (HJB) representation. Let $V(S, r)$ denote the value function, where the individual state is $S = (W, F, c, s)$ and the aggregate state is summarized by the short rate r.

$$\begin{aligned}\delta V(r,S) = \sup_{C\geq 0} \Big\{ & u(C) + \mathcal{L}_r V(r,S) + \mathcal{L}_y V(r,S) + \mu_W(r,S;C)\, V_W(r,S) + \mu_F(S)\, V_F(r,S) \\ & + \lambda^A(r,S)\left[\widetilde{V}^{refi}(r,S) - \Phi(s) - V(r,S)\right]_+ + \nu_m\left[V^m(r,S) - V(r,S)\right]\Big\}.\end{aligned} \tag{14}$$

Equation (14) is a continuous-time HJB in which the first line captures standard flow utility and the endogenous savings decision. Consumption C is chosen optimally subject to the household's intertemporal budget constraint implied by the wealth drift μ_W. The derivatives V_W and V_F denote marginal values with respect to liquid wealth W and mortgage balance F, respectively, and $\mathcal{L}_r$ and $\mathcal{L}_y$ capture the effects of aggregate interest-rate dynamics and idiosyncratic income risk on continuation values.

The second line reflects endogenous refinancing opportunities. Refinancing consideration arrives according to a Poisson process with attention intensity $\lambda^A(r, S)$, where $g(r, S) \equiv c - m(r)$ is the rate gap and $\lambda^A(r, S)$ is shorthand for $\lambda^A(g(r, S), s)$. Upon such an arrival, households compare the continuation value from refinancing to the value of remaining in the current mortgage contract; the execution decision is governed by a hassle-cost threshold, so refinancing occurs only when the value gain exceeds the income-state–dependent threshold $\Phi(s)$.

The final term captures exogenous refinancing events, such as relocation, which arrive with constant intensity ν_m and mechanically reset the mortgage coupon to the prevailing market rate.

The cash-on-hand drift and mortgage-balance drift are

$$\mu_W(r,S;C) = Y(y) - C + rW - cF - \alpha F, \qquad \mu_F(S) = -\alpha F. \tag{15}$$

The coupon-reset (exogenous refinancing) jump value is

$$V^m(r,S) = V\big(r,\ (W,F,m(r),y)\big)\,, \tag{16}$$

and the value of refinancing with optimal cash-out is

$$\widetilde{V}^{refi}(r,S) \equiv \sup_{\Delta F^{co}\in\mathcal{D}(r,S)} V^{refi}\big(r,\ (W+\Delta F^{co}-\kappa,\ F+\Delta F^{co},\ m(r),\ y)\big)\,. \tag{17}$$

The infinitesimal generators for the short rate r (CIR) and idiosyncratic income y are

$$\mathcal{L}_r V \equiv \kappa_r(\bar{r}-r)\,V_r + \tfrac{1}{2}\sigma_r^2\, r\, V_{rr}, \qquad \mathcal{L}_y V \equiv \mu_y(y)\,V_y + \tfrac{1}{2}\sigma_y^2\, V_{yy}. \tag{18}$$

Distribution dynamics. Let $g_t(r,S)$ denote the cross-sectional density of households over individual states $S=(W,F,c,y)$ and the aggregate short rate r. The evolution of g_t is governed by a Kolmogorov Forward Equation (KFE) that combines continuous drift in (W,F,r,y) with jump transitions generated by refinancing and coupon resets. Define the effective endogenous refinancing intensity as $\lambda^{refi}(r,S) \equiv \lambda^A(r,S)\,\rho(r,S), \qquad \rho(r,S)\in\{0,1\}$, where $\rho(r,S)$ is the household's refinancing policy (equal to one when the value gain exceeds the threshold and the household refinances upon an attention arrival). Then $g_t(r,S)$ satisfies

$$\begin{aligned}\partial_t g_t(r,S) \;=\; &-\partial_W\big(\mu_W(r,S), g_t(r,S)\big) - \partial_F\big(\mu_F(S), g_t(r,S)\big) + \mathcal{L}_r g_t(r,S) + \mathcal{L}_y g_t(r,S)\\ &-\big(\nu_m + \lambda^{refi}(r,S)\big)g_t(r,S) + \nu_m g_t\big(r,(\mathcal{T}^m)^{-1}(S)\big)\\ &+\lambda^{refi}\big(r,(\mathcal{T}^{refi})^{-1}(S)\big)g_t\big(r,(\mathcal{T}^{refi})^{-1}(S)\big)\end{aligned} \tag{19}$$

where $\mu_W(r,S) = Y(y) - C(r,S) + rW - cF - \alpha F$ and $\mu_F(S) = -\alpha F$ are the endogenous drifts in liquid wealth and mortgage balance. The operators $\mathcal{L}_r^*$ and $\mathcal{L}_y^*$ are the adjoints of the r- and y-generators and capture diffusion-driven mass transport in those dimensions:

$$\mathcal{L}_r^* g \equiv -\partial_r\big(\kappa_r(\bar{r} - r)\, g\big) + \tfrac{1}{2}\,\partial_{rr}\big(\sigma_r^2\, r\, g\big), \qquad \mathcal{L}_y^* g \equiv -\partial_y\big(\mu_y(y)\, g\big) + \tfrac{1}{2}\,\partial_{yy}\big(\sigma_y^2\, g\big) \tag{20}$$

The first line of (19) captures continuous evolution of the distribution due to saving/consumption choices, amortization, and the stochastic dynamics of r and y. The second line implements jump transitions. Mass leaves state (r,S) at total intensity $\nu_m + \lambda^{refi}(r,S)$ (outflow), and enters from pre-jump states that map into S through the coupon-reset mapping $\mathcal{T}^m$ and the refinancing mapping $\mathcal{T}^{refi}$ (inflow). In practice, for the mappings used in the numerical implementation, the Jacobian terms are equal to one and are omitted.

In computation, I solve the HJB and KFE jointly using an implicit finite-difference scheme with up-winding for the drift terms and sparse transition matrices for the jump components.

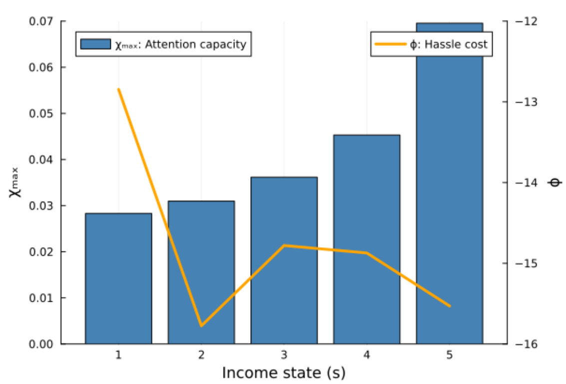
5 Quantitative Analysis

5.1 Calibration

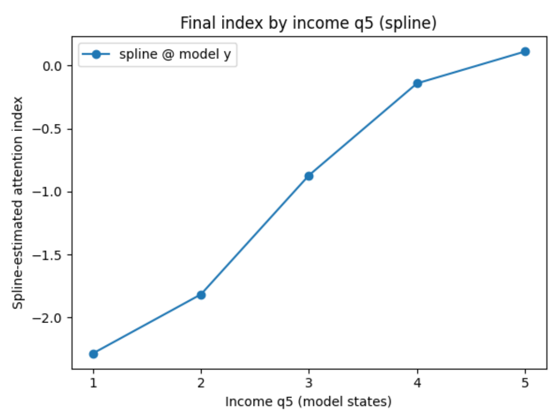
Attention and hassle cost parameters. The attention friction is governed by the parameters $\chi_{\max}(s)$, k, and μ. In the baseline calibration, I treat $\chi_{\max}(s)$ and the income-specific hassle-cost thresholds $\Phi(s)$ as a joint block of ten friction parameters and estimate them simultaneously from the refinancing hazard by income quintile. This approach allows the data to determine how much of the cross-sectional variation in hazards is absorbed by attention capacity versus execution costs, rather than imposing a priori structure on either channel.

The estimated attention capacities display a clear income gradient. The function $\chi_{\max}(s)$ rises monotonically with the income state, implying that higher-income households have a higher ceiling for how often they pay attention to refinancing opportunities when they are deep in-the-money. To provide external evidence for this pattern, I use household-level proxy measures constructed from the National Survey of Mortgage Originations (NSMO). The survey reports (i) the number of lenders a borrower considered and (ii) the number of lenders to which the borrower applied at origination. These variables capture the intensity of search and information acquisition during mortgage choice. I normalize the responses and aggregate them into a single attention index by income group. Figure 6 compares this NSMO-based index with the model-implied $\chi_{\max}(s)$ and shows that both series increase systematically with income, lending support to the interpretation of $\chi_{\max}(s)$ as an income-dependent attention capacity.

By contrast, the estimated hassle-cost thresholds $\Phi(s)$ are not monotone in income. In particular, the second income quintile exhibits a markedly lower threshold than both the bottom and middle groups. Figure 6a plots the estimated $\Phi(s)$ across quintiles. A useful way to interpret this pattern is that households in the lower-middle of the income distribution sit at the intersection of two opposing forces. On the one hand, they remain relatively liquidity constrained and have strong incentives to refinance and extract equity when opportunities arise. On the other hand, they must still cover the upfront refinancing cost κ, which limits refinancing for the very lowest-income borrowers. A lower effective hassle cost for the second quintile helps the model reconcile these forces and match the observed refinancing behavior in this part of the distribution.



(a) Estimated friction parameters



(b) NSMO-based attention index (normalized relative to quintile 3)

Figure 6: Friction parameters and NSMO attention index by income quintile

Gap sensitivity parameters. The remaining attention parameters, k and μ, are common across income states and govern the shape of the response to the rate gap. The common-shape restriction is motivated by reduced-form evidence from the data. Across income groups, the increase in refinancing activity as the rate gap widens exhibits similar curvature and timing, with differences across groups primarily reflected in the maximum refinancing rates rather than in the responsiveness to the gap itself. Allowing $\chi_{\max}(s)$ to vary by income state while keeping k and μ common is therefore sufficient to capture the observed heterogeneity in refinancing behavior.

I estimate k and μ using Nonlinear Least Squares (NLS). Specifically, for each income quintile, I fit the gap-dependent attention specification implied by equation (9) to the empirical refinancing hazard as a function of the rate gap. The estimation jointly disciplines the slope and location of the attention response while allowing the level to differ across income groups through $\chi_{\max}(s)$. This approach ensures that the calibrated attention function matches the observed shape of refinancing responses to rate gaps without overfitting income-specific sensitivities.

Other model parameters The other model parameters follows the quantitative macro-finance literature. The coefficient of relative risk aversion is set to $\gamma = 2$, a standard value in macroeconomic models. The baseline mortgage balance F, the time-preference parameter δ, and the parameters governing the interest rate process — $(\eta_r, \sigma_r, \bar{r})$ — are calibrated to match Berger et al. (2021) , implying a mortgage balance of $150,000, an annual mean short rate of 3.5%, and an interest-rate volatility of 6%. The stochastic income process follows Floden and Lindé (2001), with a log-income persistence corresponding to a half-life of 7.3 years and an annualized volatility of 21%.

The payment-to-income ratio $\phi^{PTI} = 0.43$ limits the maximum feasible mortgage payment relative to income, while the loan-to-value ratio $\phi^{LTV} = 0.8$ restricts the maximum outstanding balance relative to home value. Both parameters are calibrated to match average underwriting standards in the Freddie Mac.

Further details on the cross-sectional distribution of the baseline model are provided in Figure A.3.

Table 5: Model Parameter Values

Panel A: Exogenous Processes			
Parameter	**Value**	**Description**	**Source**
$\ln 2/\eta_r$	5.3 years	Half-life of interest-rate shock	Berger et al. (2021)
$\bar{r}$	3.5% (p.a.)	Unconditional mean short rate	Berger et al. (2021)
σ_r	6% (p.a.)	Volatility of short rate	Berger et al. (2021)
$\ln 2/\eta_y$	7.3 years	Half-life of (log) income shock	Flodén and Lindé (2001)
$\mathbb{E}[Y_t]$	\$58,000	Unconditional mean income	Flodén and Lindé (2001)
σ_y	21% (p.a.)	Log-income volatility	Flodén and Lindé (2001)
γ	2	Inverse of IES	Macro literature standard
F	\$150,000	Mortgage balance outstanding	Berger et al. (2021)

Panel B: Refinancing Frictions			
Parameter	**Hybrid**	**Description**	**Source**
ν_m	4.1% (p.a.)	Arrival rate of exogenous moving shocks	Berger et al. (2021)
$\chi_{\max}(s)$	{0.028, 0.031, 0.036, 0.045, 0.069}	Maximum attention capacity by income quintile	Estimated
$\log \Phi(s)$	{−12.8, −15.8, −14.8, −14.8, −15.5}	Maximum attention capacity by income quintile	Estimated
k	1.931	Responsiveness of attention to rate gap	Estimated
μ	0.284	Location parameter of attention response to rate gap	Estimated
κ	\$3,000	Fixed (menu) cost of refinancing	Eichenbaum et al. (2021)
ϕ^{PTI}	0.43	Payment-to-income constraint ratio	Freddie Mac
ϕ^{LTV}	0.80	Maximum loan-to-value ratio	Freddie Mac

5.2 Targeted Hazard Moments

I first evaluate the model against the moments that discipline the refinancing-friction parameters. The targeted moments are monthly refinancing hazards by interest-rate-gap bins and income quintile. These moments are central to the quantitative exercise because they jointly identify the two dimensions of heterogeneity emphasized in the model. Variation across rate-gap bins disciplines how strongly and at what point attention responds to rate gaps, summarized by k and μ. Variation across income quintiles disciplines the income-dependent components of refinancing frictions: the maximum attention capacity, $\chi_{\max}(s)$, and the hassle-cost threshold, $\Phi(s)$.

The empirical counterpart is the refinancing hazard by rate-gap bin and income quintile estimated from the loan-level panel, using the same rate-gap measure as in the reduced-form analysis shown in Figure 3. The model counterpart is constructed by initializing the simulated mortgage distribution with the income-specific distribution of outstanding coupons observed in the loan panel in January 2005.[11] Holding this initial distribution fixed, I feed in the realized path of mar-

[11]The empirical sample begins in 2000, but the model simulation starts from the mid-sample to allow for a sufficient burn-in period.

ket mortgage rates and compute monthly refinancing hazards within the same rate-gap bins and income groups used in the empirical analysis.

Figure 7 compares the model-implied hazards with their empirical counterparts. The baseline model reproduces the main patterns in the data. Refinancing hazards rise with the rate gap, but they remain far below one even for borrowers who are deep in the money. This feature is informative about the attention friction: profitable refinancing opportunities are not acted upon mechanically, even when the monetary gains are large. The model also captures the income gradient in refinancing hazards, with substantially higher refinancing propensities among borrowers in the fourth and fifth income quintiles than among those in the lower part of the distribution.

Overall, the targeted moments show that the model captures both margins of refinancing behavior that matter for monetary transmission: responsiveness to rate gaps and heterogeneity in that responsiveness across the income distribution. These moments therefore provide the quantitative discipline for the counterfactual exercises below.

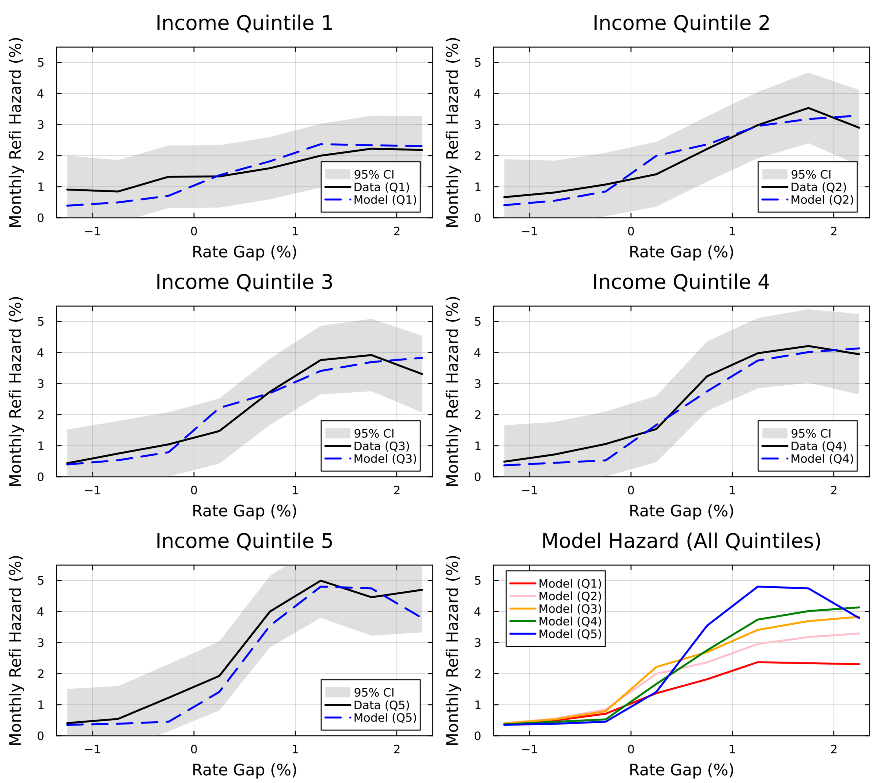


Figure 7: Hazard Rate by Gap Bin: Model vs. Data

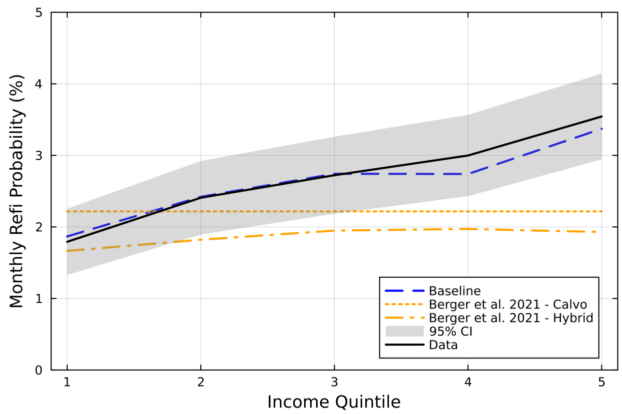
5.3 Untargeted Model Validation

Average refinancing hazards. As an untargeted cross-sectional validation, I next examine the average monthly refinancing probability among *in-the-money* borrowers by income quintile. This moment is not directly targeted in the calibration, which is disciplined primarily by refinancing hazards across rate-gap bins. It is nevertheless informative because it summarizes whether the calibrated model reproduces the aggregate income gradient in refinancing behavior that motivates the paper.

Figure 8 shows that refinancing propensity in the data increases monotonically with income, producing a clear upward-sloping income gradient across quintiles. The baseline model with income-dependent frictions successfully captures this pattern. In contrast, a Calvo-style attention-only benchmark with common parameters across households generates an essentially flat profile. Because this benchmark is calibrated in Berger et al. (2021) using different data sources and a different sample period, its overall level is not directly comparable to the empirical hazards in my loan panel. The relevant comparison is therefore the cross-sectional shape: the benchmark does not generate the upward-sloping income gradient observed in the data. A hybrid benchmark that allows for additional channels beyond pure attention produces only a muted gradient, reflecting the absence of sufficiently strong income-dependent refinancing frictions. Taken together, these comparisons show that matching the average level of refinancing is not sufficient. Without heterogeneity in refinancing frictions across income groups, the model cannot replicate the inequality in refinancing propensity that underlies heterogeneous monetary transmission.

Distributional moments. I next examine whether the model delivers plausible cross-sectional heterogeneity in consumption, liquid wealth, and marginal propensities to consume. These moments are not targeted in the calibration, but they are important for interpreting the monetary-transmission results because refinancing affects aggregate consumption through household cash flows and MPC heterogeneity.

Figure 9 reports the Lorenz curves for consumption and liquid wealth to their empirical counterparts from the CEX (1988–2018) and the 2001 SCF, and I contrast the model-implied MPC

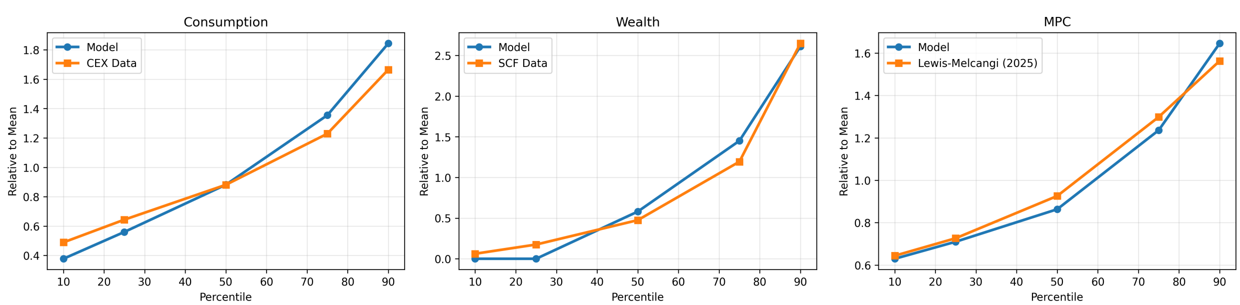


(a) Baseline vs Berger et al. (2021)

Figure 8: Average Hazard Rate by Income Quintile: Model vs. Data

profile with the estimates in Lewis et al. (2025). All objects are computed at percentiles of the cross-sectional distribution and normalized by their means, so the comparison emphasizes shapes rather than levels. The model reproduces the broad distributional patterns in the data—realistic inequality in liquid wealth and consumption—and generates an MPC profile that is close to the empirical benchmark.

This validation matters because the quantitative transmission results depend on where refinancing gains accrue in the liquidity distribution. The fact that the model generates plausible liquid-wealth inequality and MPC heterogeneity reduces the concern that the consumption IRFs and counterfactual results are driven by an unrealistic distribution of liquidity or marginal propensities to consume.



(a) Consumption (b) Liquid Wealth (c) MPC

Figure 9: Lorenz Curve and MPC: Model vs. Data

Average coupon dynamics. The Figure 10, compares the evolution of average mortgage coupon rates in the model to the data. In the empirical series I focus on outstanding mortgages, exclude purchase originations, and compute the average coupon rate over the remaining loans. The model counterpart is constructed in the same way, based on the simulated cross-sectional distribution of contracts. For reference, I also plot the 30-year fixed mortgage rate on new originations, which does not coincide with the average coupon on outstanding debt but traces the underlying interest rate path. The model starts from a similar initial level and tracks the subsequent decline in the average coupon rate, although the adjustment is somewhat slower. This residual gap appears to be related to differences between the model-implied distribution of borrowers across income groups and the distribution observed in the data.

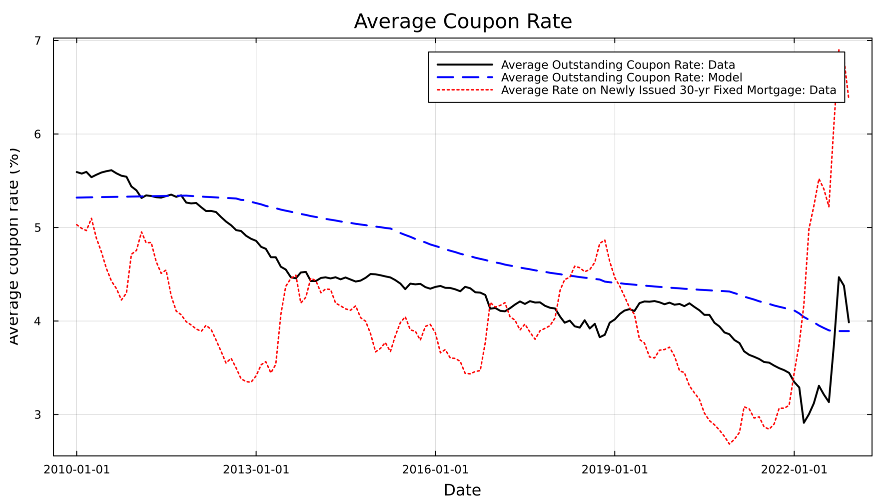


Figure 10: Average Coupon of Outstanding Loans: Model vs. Data

5.4 Counterfactual Experiments

Homogeneous frictions. I first ask whether the model can reproduce the observed income gradient without income-dependent refinancing frictions. In this counterfactual, I restrict both friction parameters to be common across income states, setting $\chi_{\max}(s) = \bar{\chi}$ and $\Phi(s) = \bar{\Phi}$ for all s, and re-calibrate the common parameters to match overall refinancing activity.

Figure 11 compares the resulting hazard schedules with the baseline model and the data. The homogeneous-frictions model preserves the basic upward relationship between refinancing hazards and the rate gap, but it compresses the cross-income differences in those hazards. It tends to generate too much refinancing among lower-income borrowers and too little refinancing among higher-income borrowers, especially in the in-the-money region where the empirical income gradient is most pronounced.

Figure 12 summarizes the same failure in average in-the-money refinancing rates by income quintile. While the data and the baseline model display a clear upward-sloping income gradient, the homogeneous-frictions counterfactual is nearly flat across income groups. Thus, matching the average level of refinancing activity is not sufficient. The model needs income-dependent frictions to match who refinances when interest rates fall, which is the margin that matters for heterogeneous monetary transmission.

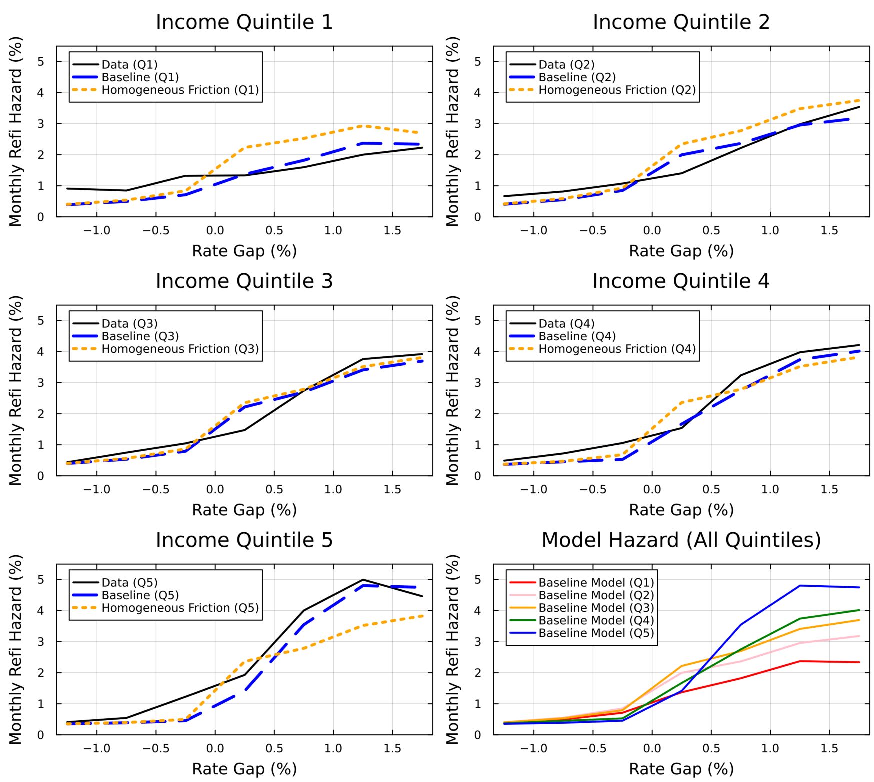


Figure 11: Hazard Rate by Gap Bin: Counterfactual Homogeneous Frictions

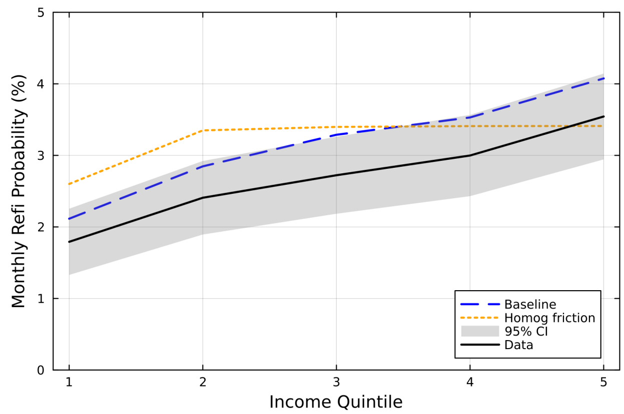


Figure 12: Average Hazard Rate: Counterfactual Homogeneous Frictions

Friction channels and heterogeneity To assess how much each refinancing friction and its income heterogeneity contribute to the model fit, I run a set of counterfactual experiments that selectively remove individual channels. In the first experiment, shown in Figure 13, I shut down the hassle-cost channel and retain only the gap-dependent attention friction, recalibrating the remaining five attention parameters by income group. The hazard curves for the bottom four quintiles remain close to the baseline, but the model now understates refinancing activity for the top quintile, especially at larger rate gaps. The intuition is that removing the execution threshold raises the continuation value for high-income households that expect to move down the income distribution in the future. Because refinancing becomes more attractive in lower income states, the expected gain from refinancing while still in quintile 5 falls, which depresses current hazards.

Figure 14 instead shuts down attention frictions and lets refinancing be determined entirely by the policy function implied by hassle costs. In this case the model resembles a standard menu-cost environment: refinancing probabilities are near zero for small gaps and jump steeply once the threshold is crossed. This specification cannot match the empirical feature that refinancing rates remain well below one even for deep in-the-money borrowers. Hence a pure menu-cost formulation cannot generate the refinancing inequality observed in the data.

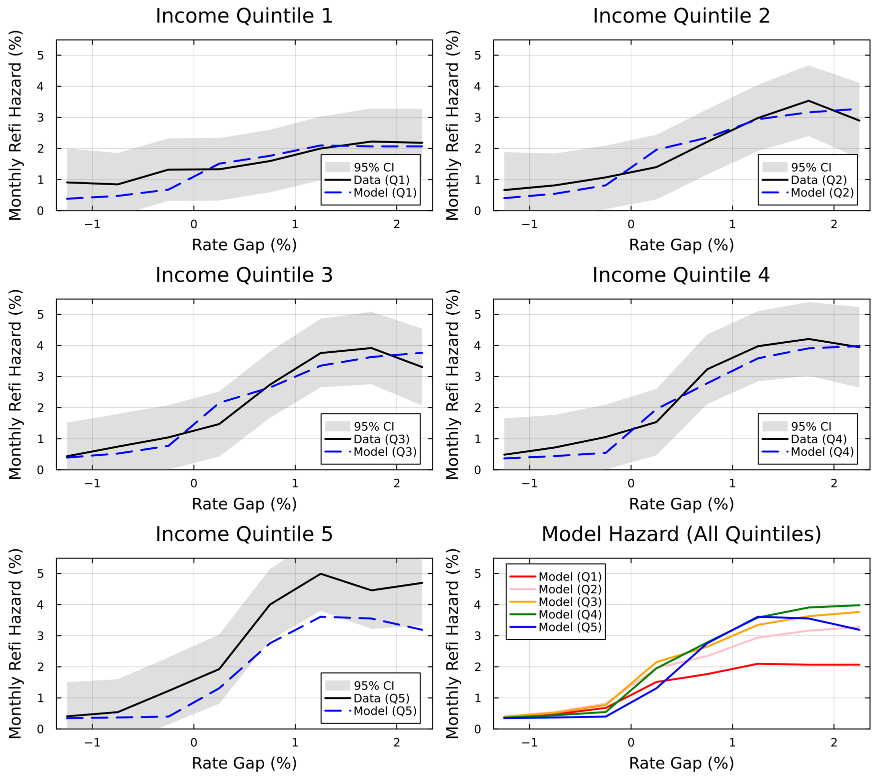


Figure 13: Hazard Rate by Gap Bin: Attention Friction Only

5.5 Consumption Response to Monetary Policy

I now quantify how unequal refinancing responses shape the consumption effects of monetary policy. A decline in interest rates raises the value of refinancing for borrowers with high outstanding coupons. The frictions defined in Section 4 determine how many borrowers respond, how quickly they respond, and where the resulting cash-flow relief accrues in the income and liquidity distribution. The consumption impulse response therefore summarizes the interaction between the interest-rate stimulus, refinancing frictions, cash-out choices, and household MPC heterogeneity.

IRF construction. I compute the response to a one-percentage-point decline in the policy rate by comparing two economies that start from the same cross-sectional distribution of liquid wealth, mortgage balances, coupon rates, and income.[12] In the baseline path, the economy evolves without

[12]The exercise is partial equilibrium: the policy-rate path is taken as given, and I do not model feedback from aggregate consumption to interest rates, house prices, or lender balance sheets.

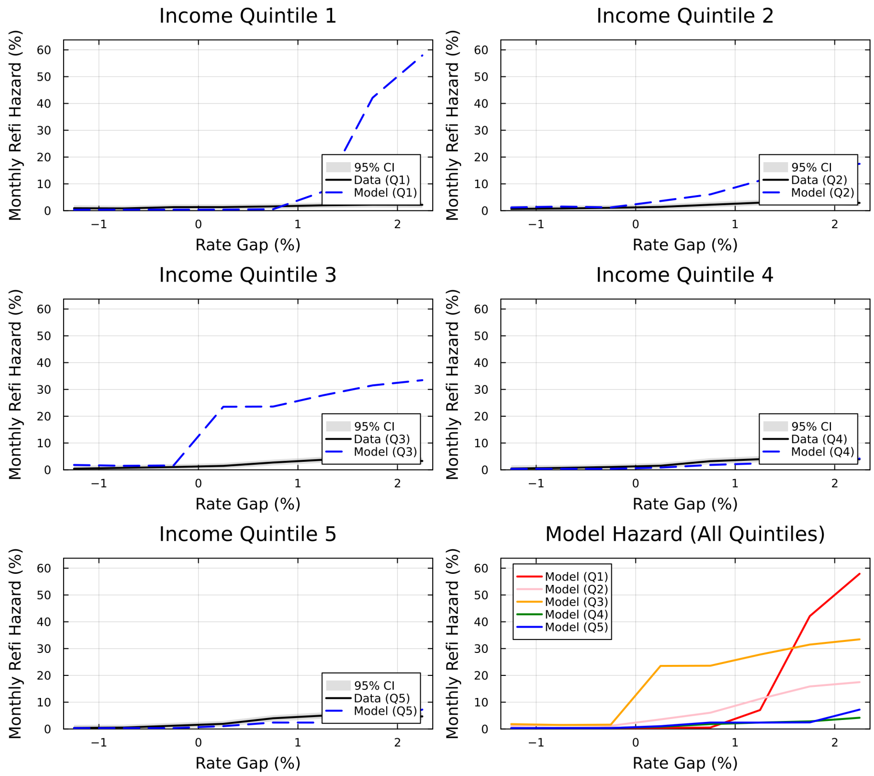
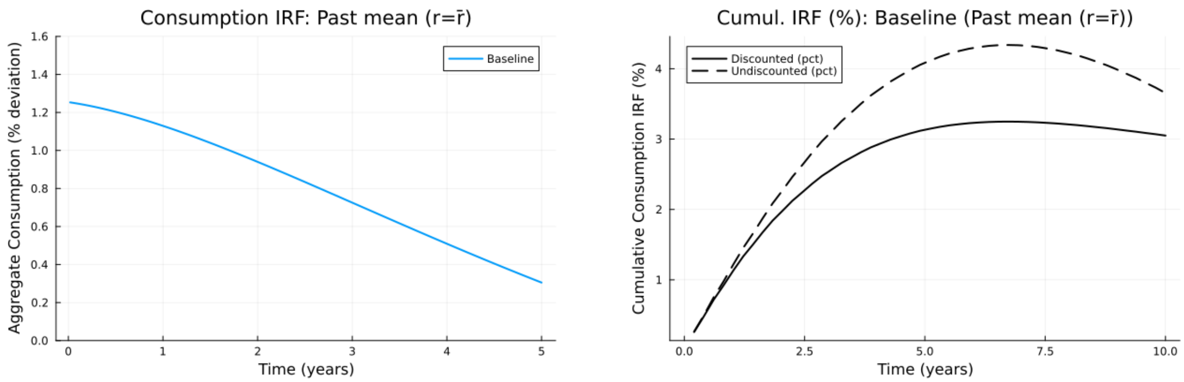


Figure 14: Hazard Rate by Gap Bin: Hassle Cost Only

the rate cut. In the shocked path, the policy rate is lowered at time zero and the distribution is simulated forward in the same model environment. Aggregate consumption is computed by integrating the household consumption policy over the evolving distribution. The consumption IRF is measured as the percentage deviation of aggregate consumption in the shocked economy from the baseline economy at each horizon. Details of the IRF construction are provided in Appendix E.

Baseline transmission. I first examine the baseline economy, which keeps the estimated income-dependent refinancing frictions unchanged. This exercise establishes the benchmark strength of the refinancing channel before considering any counterfactual reduction in frictions. Figure 15 shows the aggregate consumption response to a 100 basis-point decline in the policy rate. The response is sizable on impact and remains positive over the subsequent years. Over the first five years, aggregate consumption is on average 0.82 percent above the baseline path.[13] The relatively front-loaded response reflects the experiment's focus on the refinancing channel: the rate cut immediately changes rate gaps and household policy functions, while the subsequent dynamics are governed by the stochastic arrival of refinancing opportunities, cash-out decisions, and the evolution of the mortgage distribution.



(a) Consumption IRF

(b) Cumulative Consumption IRF

Figure 15: Consumption Response to a 1 pp Rate Cut

Reduced-frictions counterfactual. I next study a reduced-frictions counterfactual disciplined by direct experimental evidence. Byrne et al. (2023) conduct a field experiment in which eligible mortgage borrowers received redesigned refinancing disclosures and reminder notices. Their experiment shows that lowering informational and attentional barriers substantially raises refinancing take-up: in the benchmark treatment with reminders, the refinancing rate rises from 8.9 percent in the control group to 15.7 percent, implying an increase of roughly 60–75 percent depending on the treatment specification. The estimated effects are therefore most naturally interpreted as an increase in borrowers' attention to refinancing opportunities, rather than as a change in the financial gains from refinancing or in the mortgage contract environment.

I adapt this baseline experimental estimate to a market-wide reminder intervention. Specifically, I increase the attention-capacity parameters of all income quintiles by 60 percent, leaving the

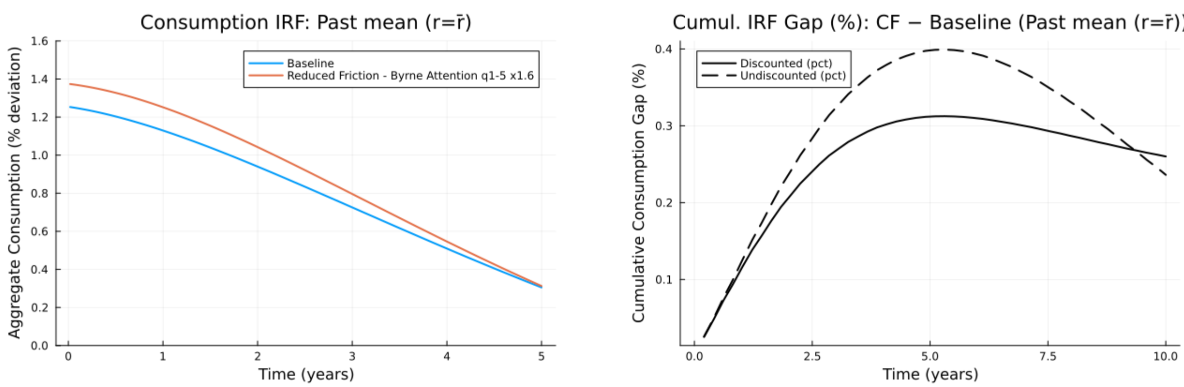
[13] The five-year average response is computed as $\frac{1}{5}\int_0^5 IRF_C(t)\,dt$. In the simulation, $\int_0^5 IRF_C(t)\,dt = 4.085$ percentage-point-years, implying an average response of $4.085/5 = 0.817\%$. The impact response is 1.25 percent. I interpret this number as the partial-equilibrium response of the refinancing cash-flow channel, rather than as a full aggregate monetary-policy multiplier.

execution-cost parameters and all other primitives unchanged. Formally, I set

$$\chi^{cf}_{\max}(s) = 1.6\chi_{\max}(s), \qquad s \in \{1, \ldots, 5\},$$

and keep $\Phi^{cf}(s) = \Phi(s)$ for all income groups. The experiment can be interpreted as a scalable notice policy that operates through attention frictions in refinancing.

Figure 16 shows that the counterfactual amplifies monetary transmission. Over the first five years after a 100 basis point policy-rate cut, the discounted cumulative consumption response rises from 3.13 percentage points in the baseline economy to 3.44 percentage points under the Byrne-scaled attention counterfactual. The difference, 0.31 percentage points, represents a 10.0 percent increase relative to the baseline cumulative response. Thus, a broad reduction in attention frictions generates a meaningful increase in the consumption response to monetary policy.



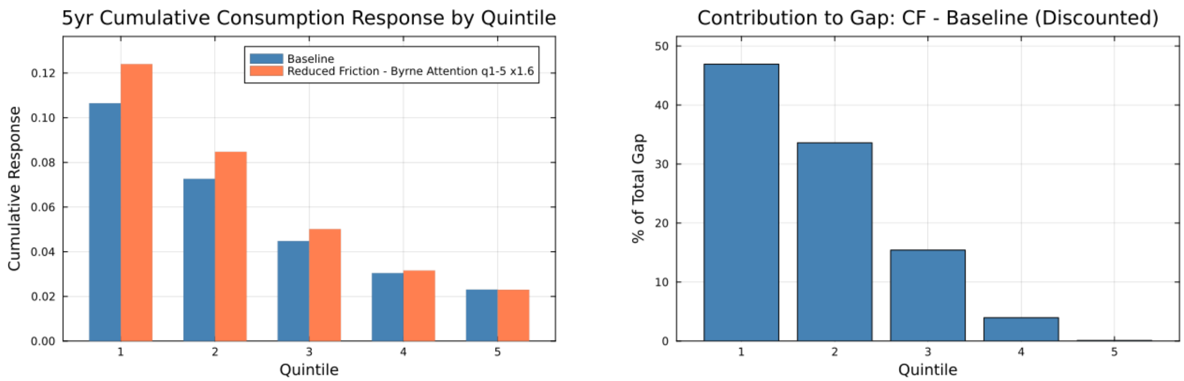
(a) Consumption IRF

(b) Cumulative Consumption IRF

Figure 16: Consumption Response to a 1 pp Rate Cut: Byrne-Scaled Attention Counterfactual

Notes: The figure compares the baseline economy with a reduced-frictions counterfactual in which the attention-capacity parameters $\chi_{\max}(s)$ of all income quintiles are increased by 60 percent, consistent with the refinancing response in Byrne et al. (2023). Panel (a) reports the aggregate consumption IRF to a 1 percentage point rate cut. Panel (b) reports the cumulative gap between the counterfactual and baseline responses.

To understand where this amplification comes from, I decompose the additional five-year cumulative consumption response by income quintile. Figure 17 shows that the extra response is concentrated at the bottom of the income distribution even though the attention intervention is applied to all borrowers. The bottom two quintiles account for about 80.5 percent of the discounted increase in the five-year cumulative consumption response, with 46.9 percent coming from the bottom quintile and 33.6 percent from the second quintile. By contrast, the top quintile contributes essentially none of the additional response. This pattern reflects the joint distribution of refinancing gains, attention frictions, and marginal propensities to consume: raising attention increases refinancing opportunities throughout the distribution, but the resulting cash-flow relief has the largest spending effects when it reaches lower-income, higher-MPC households.



(a) Cumulative response by quintile

(b) Contribution to the additional response

Figure 17: Distributional Decomposition of the Byrne-Scaled Counterfactual

Notes: Panel (a) reports the five-year cumulative consumption response by income quintile in the baseline economy and in the market-wide Byrne-scaled attention counterfactual. Panel (b) reports each quintile's share of the discounted counterfactual-minus-baseline increase in the five-year cumulative response.

I also examine two additional reduced-friction scenarios, reported in Appendix F.1. First, I consider a targeted reminder policy that raises attention only among the bottom two income quintiles, setting $\chi^{cf}_{\max}(s) = 1.6\chi_{\max}(s)$ for $s \in \{1, 2\}$ and leaving all other income groups unchanged. This more conservative intervention increases the five-year discounted cumulative consumption response by 6.2 percent relative to baseline. Second, I consider a more aggressive equalized-frictions counterfactual in which all households face the refinancing-friction parameters of the top income quintile, $\chi^{cf}_{\max}(s) = \chi_{\max}(5)$ and $\Phi^{cf}(s) = \Phi(5)$ for all s. In that case, the five-year discounted cumulative consumption response rises by 13.7 percent relative to baseline. These exercises show that the main result is not specific to a single counterfactual design: weaker refinancing frictions consistently amplify monetary transmission, with larger effects when the reduction in frictions is broader or more intensive.

MPC sensitivity. The untargeted validation above shows that the model delivers a plausible MPC profile relative to the empirical estimates in Lewis et al. (2025). Nevertheless, because the transmission results depend on how refinancing-induced cash-flow relief is converted into spending, I examine the sensitivity of the amplification to the income gradient in MPCs. Let m_s denote the model-implied average MPC for income state s, and let $\bar{m}$ be its cross-sectional average. I construct alternative MPC schedules

$$m_s(\lambda) = \bar{m} + \lambda(m_s - \bar{m}), \qquad \lambda \in \{0, 0.5, 1, 1.5\},$$

holding fixed the model-implied refinancing and payment-relief paths. The case $\lambda = 0$ removes the income gradient in MPCs, $\lambda = 1$ preserves the model-implied gradient, and $\lambda = 1.5$ steepens it. For each schedule, I compute an accounting consumption response by applying $m_s(\lambda)$ to the payment relief received by each income group.

In the structural model, the market-wide Byrne-scaled attention counterfactual raises the five-year discounted cumulative response by 10.0 percent relative to baseline. The corresponding accounting exercises imply discounted amplifications between 3.4 and 4.1 percent. Thus, the exact magnitude depends on the strength of the cash-flow-to-consumption pass-through, but the qualitative result is stable: reducing attention frictions raises monetary transmission, and the additional response remains concentrated among lower-income borrowers. I report the detailed sensitivity exercises in Appendix F.2.

Policy implications. Because the counterfactual is disciplined by field-experimental evidence, it provides a direct bridge from the model to policy. Reminder-style interventions, mandatory in-the-money notices, and other information policies that make refinancing opportunities more salient correspond in the model to higher attention capacity. The structural framework translates such interventions into a common outcome metric: how much they amplify the aggregate consumption response to monetary policy. The exercise also shows why the distributional incidence of these policies matters. Even when attention is raised market-wide, the additional spending response is concentrated among lower-income households, for whom refinancing-induced cash-flow relief is more likely to translate into consumption. Thus, policies that reduce attention frictions can both strengthen monetary transmission and redirect its cash-flow channel toward households with higher marginal propensities to consume.

5.6 Implications on Inequality

Finally, I examine how reduced refinancing frictions change long-run inequality in the stationary distribution. Table 6 reports two counterfactuals. The first is the evidence-based attention counterfactual used as the main policy experiment above, in which attention capacity rises by 60 percent for all income states. The second is a more aggressive friction-reduction exercise in which all income groups face the attention and hassle-cost parameters of the top income quintile.

The evidence-based attention counterfactual produces little movement in consumption inequality: the consumption Gini remains at 0.30, the top-10 consumption share is essentially unchanged, and the bottom-50 consumption share moves only slightly from 28.42% to 28.35%. The effect on liquid-wealth inequality is more visible, although still moderate in magnitude. The liquid-wealth Gini declines from 0.56 to 0.54, the top-10 liquid-wealth share falls from 33.05% to 32.09%, and the bottom-50 liquid-wealth share rises from 9.10% to 10.60%. Thus, a policy-sized reduction in attention frictions is sufficient to generate a visible compression in liquidity positions, but it does not materially alter the long-run distribution of consumption.

The more aggressive top-quintile-frictions counterfactual shows that larger reductions in refinancing frictions can generate more meaningful stationary wealth effects. In that case, the liquid-wealth Gini falls further to 0.53, the top-10 liquid-wealth share declines to 31.46%, and the bottom-50 share rises to 11.41%. Consumption inequality remains nearly unchanged even under this larger friction reduction. These comparisons suggest that refinancing frictions primarily affect long-run inequality through the distribution of liquid assets rather than through permanent differences in consumption. The mechanism is consistent with refinancing easing household liquidity constraints and allowing lower-wealth households to retain more cash flow in the stationary distribution.

Table 6: Stationary Inequality under Reduced Refinancing Frictions

	Consumption				Liquid wealth			
Metric	Base	CF	CF–Base	%Δ	Base	CF	CF–Base	%Δ
Evidence-based attention								
Gini	0.30	0.30	0.00	0.33%	0.56	0.54	-0.02	-3.67%
Top 10 share	20.62%	20.62%	0.00 pp	0.02%	33.05%	32.09%	-0.96 pp	-2.91%
Bottom 50 share	28.42%	28.35%	-0.07 pp	-0.25%	9.10%	10.60%	1.50 pp	16.53%
Top-quintile frictions								
Gini	0.30	0.30	0.00	0.44%	0.56	0.53	-0.03	-5.47%
Top 10 share	20.62%	20.62%	0.00 pp	0.01%	33.05%	31.46%	-1.59 pp	-4.82%
Bottom 50 share	28.42%	28.31%	-0.11 pp	-0.37%	9.10%	11.41%	2.32 pp	25.47%

Notes: This table compares stationary-distribution inequality in the baseline economy and two reduced-frictions counterfactuals. In the evidence-based attention counterfactual, $\chi_{\max}(s)$ is increased by 60 percent for all income states. In the top-quintile frictions counterfactual, all income states are assigned the attention and hassle-cost parameters of the top income quintile. Share measures are reported in percent, and CF–Base differences for share measures are reported in percentage points.

6 Conclusion

This paper examines how refinancing frictions are distributed across the income distribution and how that distribution shapes the transmission of monetary policy through household mortgage cash flows. Using the Freddie Mac Single-Family Loan-Level Dataset covering loans originated between 2000 and 2022, I show that refinancing inequality is large in both the *intensity* and the *timing* of refinancing. Once households become financially advantageous to refinance, the bottom income quintile refinances at roughly two-thirds the monthly rate of the top quintile, and reaches an 80% refinancing survival threshold approximately six months later than top-quintile borrowers. The unrealized refinancing savings concentrated among lower-income households exceed 7.6% of monthly income — a persistent and recurring cash-flow shortfall over the remaining life of the loan. These patterns survive controls for credit quality, updated home equity, lender-side credit tightness, origination lender-by-year fixed effects identified via the HMDA link, and income-mobility checks using matched refinancing income and the PSID.

Beyond documenting these empirical patterns, I develop a structural mortgage refinancing model that nests Berger et al. (2021) but allows two distinct refinancing frictions — an attention margin and an execution-cost margin — to vary by income state. The model delivers two findings central to the paper. First, a homogeneous-friction benchmark calibrated to the Berger et al. (2021) specification cannot account for the observed income gradient: it generates near-uniform refinancing rates across income quintiles where the data exhibit a steep and monotonic increase. Matching the empirical gradient therefore requires income-state-dependent frictions; the average level of frictions is insufficient. Second, decomposing the gradient into its attention and execution-cost components reveals that the attention channel is the dominant driver of refinancing inequality across the income distribution.

The quantitative implications of these frictions for monetary policy transmission are substantial. Guided by field-experimental evidence in Byrne et al. (2023), I evaluate a market-wide notice intervention that raises the attention-capacity parameter by 60 percent across all income states. Under this evidence-based counterfactual, the model-implied five-year discounted cumulative consumption response to a 100 basis-point policy rate cut rises from 3.13 percentage points in the baseline economy to 3.44 percentage points — an amplification of approximately 10 percent. Crucially, the additional response is highly concentrated among lower-income households: the bottom two income quintiles account for roughly 80 percent of the discounted consumption gain. This concentration reflects the joint distribution of refinancing gains, attention frictions, and marginal propensities to consume, and demonstrates that even a market-wide reduction in refinancing frictions transmits disproportionately through households for whom cash-flow relief has the largest spending effects.

These findings carry direct policy implications. Because the structural decomposition identifies inattention as the dominant friction generating refinancing inequality, the counterfactual maps onto concrete and feasible instruments — mandatory in-the-money refinancing notices, streamlined application procedures that lower the execution cost of acting on those notices, and income-targeted lender outreach. The structural framework allows such alternatives to be evaluated against a common metric: the implied amplification of the aggregate consumption response to monetary policy and the redistribution of cash-flow relief toward lower-income, higher-MPC households. Notably, the two policy goals reinforce each other — reducing attention frictions precisely where they bind most tightly simultaneously strengthens monetary transmission and compresses the cross-income distribution of mortgage payment relief.

Overall, this paper sheds light on an important yet underexplored dimension of monetary policy transmission: the income distribution of refinancing frictions. By documenting refinancing inequality in both intensity and timing, embedding income-state-dependent frictions in a structural mortgage refinancing model, and quantifying the resulting attenuation of monetary transmission, the analysis provides a framework for assessing interventions aimed at strengthening the refinancing channel of monetary policy and narrowing the cross-income distribution of its cash-flow benefits. More broadly, the results suggest that who refinances — not only whether refinancing occurs — is central to understanding how monetary policy reaches household consumption, and that the joint distribution of behavioral frictions and marginal propensities to consume merits explicit attention in the design of both monetary policy and mortgage market regulation.

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A Appendix: Tables and Figures

A.1 Figures and Tables

Table A.7: Mean Values by Income Quintile

Income Quintile	Loan ID (count)	Credit Score	LTV (%)	PTI (%)	Initial Loan Balance ($)
1	90,331	744.3	69.1	39.5	113,079.9
	<24.1>	(246.7)	(20.4)	(8.7)	(48,140.0)
2	78,577	746.9	72.8	36.9	180,094.7
	<21.0>	(201.9)	(17.5)	(9.2)	(65,313.4)
3	73,251	750.1	73.4	35.1	236,640.2
	<19.5>	(181.9)	(16.6)	(9.6)	(88,343.2)
4	68,386	756.5	73.1	32.7	299,712.5
	<18.2>	(220.8)	(16.1)	(9.7)	(113,507.7)
5	64,332	765.6	69.9	25.5	368,110.3
	<17.2>	(235.0)	(16.4)	(9.5)	(144,937.8)
Total	374,877	751.8	71.5	34.5	228,178.7
	<100.0>	(219.6)	(17.8)	(10.4)	(130,095.4)

Income Quintile	Maturity (Month)	Income ($)	Monthly Payment ($)	House Value ($)
1	328.8	18,251.7	595.0	180,890.9
	(66.1)	(5,230.7)	(205.7)	(106,661.4)
2	325.2	30,718.6	943.4	264,049.8
	(68.7)	(4,457.3)	(263.5)	(132,821.6)
3	323.6	42,484.3	1,239.8	341,449.0
	(70.0)	(5,611.8)	(366.4)	(172,616.7)
4	319.7	58,208.9	1,578.9	431,965.9
	(72.8)	(8,015.1)	(488.9)	(214,241.4)
5	301.9	103,865.1	2,026.4	562,296.5
	(82.9)	(58,277.9)	(715.0)	(305,448.9)
Total	320.8	47,369.5	1,214.4	339,725.3
	(72.3)	(38,036.0)	(654.4)	(232,158.2)

Notes: The table reports means and standard deviations (in parentheses) by income quintile. The Loan ID column represents the unique number of loans, and the values within the angle brackets indicate the proportion of loans within that quintile relative to the total number of loans. Unconditional income quintiles are divided into 20% segments each, but the matched panel is based on matched refinanced loans. Therefore, the proportion of the matched panel can vary according to the number of refinanced loans in each income quintile.

Table A.8: Interest Rate Reductions by Refinancing

Dependent	Interest Rate Reduction by Refi (%p)				
Equation	(1)	(2)	(3)	(4)	(5)
IncomeQuintile 2	-0.186[a]	0.0108[a]	0.0166[a]	0.000147	0.000926
	(0.00317)	(0.00273)	(0.00293)	(0.00298)	(0.00296)
IncomeQuintile 3	-0.294[a]	0.0134[a]	0.0216[a]	-0.00529[c]	-0.00741[b]
	(0.00320)	(0.00283)	(0.00307)	(0.00319)	(0.00317)
IncomeQuintile 4	-0.370[a]	0.0141[a]	0.0249[a]	-0.0112[a]	-0.0176[a]
	(0.00328)	(0.00297)	(0.00324)	(0.00345)	(0.00343)
IncomeQuintile 5	-0.446[a]	0.0296[a]	0.0440[a]	-0.0133[a]	-0.0198[a]
	(0.00342)	(0.00319)	(0.00350)	(0.00402)	(0.00400)
Constant	2.283[a]	0.963[a]	0.949[a]	1.059[a]	1.045[a]
	(0.00229)	(0.00701)	(0.00709)	(0.00846)	(0.00843)
Control variables					
LTV, credit score, loan age, ZIP code		x	x	x	x
Credit tightness, HARP			x	x	x
Loan balance to Income				x	x
Est. equity					x
Observations	482,078	481,683	481,683	481,683	481,683
R-squared	0.043	0.351	0.352	0.354	0.361

	Predicted Interest Rate Reduction by Refi (%p)				
Equation	(1)	(2)	(3)	(4)	(5)
IncomeQuintile 1	2.2828	2.0202	2.0147	2.0405	2.0434
	(0.0023)	(0.0021)	(0.0021)	(0.0022)	(0.0022)
IncomeQuintile 2	2.0965	2.0310	2.0319	2.0415	2.0444
	(0.0022)	(0.0019)	(0.0019)	(0.0019)	(0.0019)
IncomeQuintile 3	1.9888	2.0337	2.0353	2.0345	2.0349
	(0.0022)	(0.0019)	(0.0019)	(0.0019)	(0.0019)
IncomeQuintile 4	1.9128	2.0344	2.0370	2.0268	2.0234
	(0.0024)	(0.0020)	(0.0020)	(0.0020)	(0.0020)
IncomeQuintile 5	1.8372	2.0498	2.0524	2.0204	2.0163
	(0.0025)	(0.0022)	(0.0022)	(0.0025)	(0.0025)

Notes: This table presents estimated interest rate savings through refinancing using Equation 5. The upper section of the table displays the regression results, while the lower section shows predicted values of the dependent variable based on the regression results. Superscripts a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis.

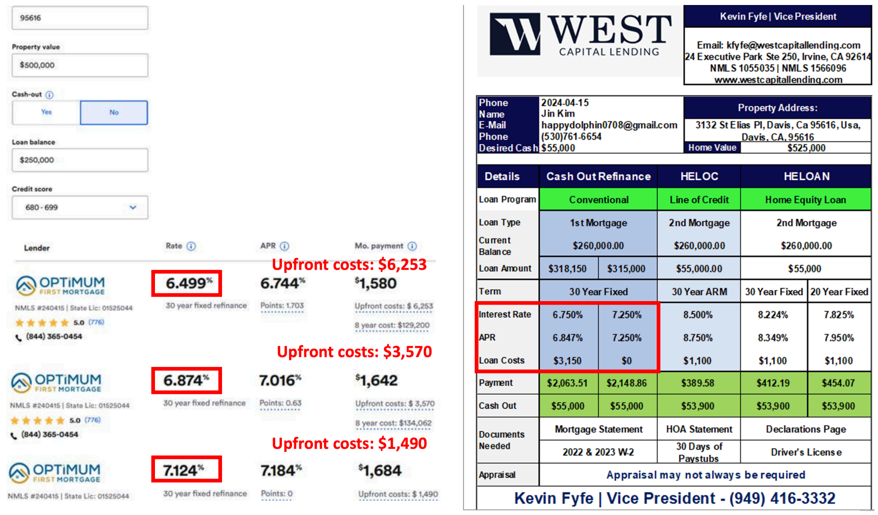


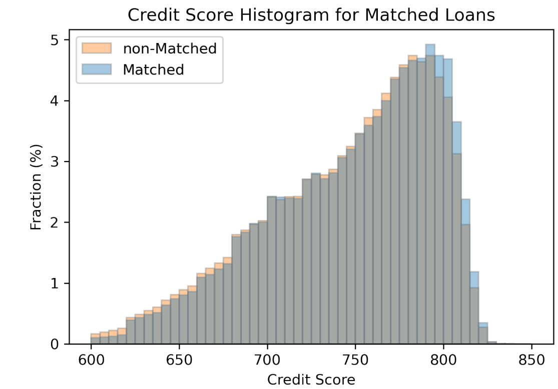
Figure A.1: Illustration of the Rate–Closing Cost Tradeoff in Mortgage Refinancing

Notes: This figure shows the offers received from mortgage brokers in a hypothetical scenario for refinancing a home in Davis, California, as of May 2024. The left figure assumes pure rate refinancing (non-cash out), while the right figure assumes a cash-out scenario. Both figures illustrate the tradeoff between upfront costs and interest rates.

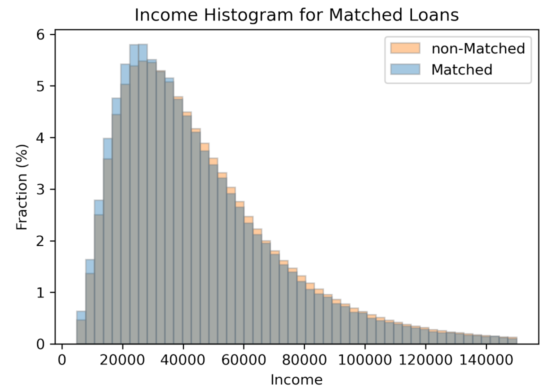
A.2 Data Validity Checks

Table A.9: Data Validity Checks on the Matched Panel

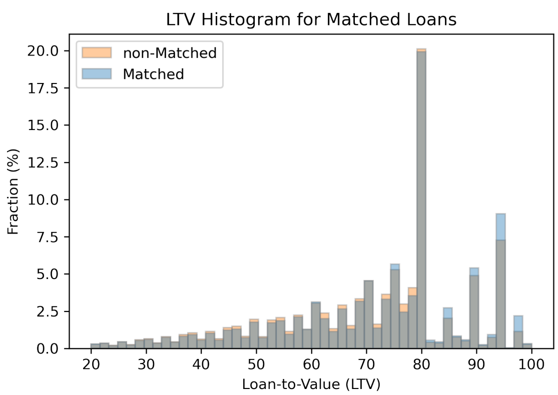
	Loan ID (count)	Credit Score	Loan Balance	Loan Term	Income	LTV	PTI
Matched Panel (mean)	374,877	751.8	228,179	320.8	47,369	71.5	34.5
All (mean)	41,150,001	758.4	213,022	309.4	49,514	70.1	33.3
Difference	*0.9% of total*	-0.9%	7.1%	3.7%	-4.3%	2.0%	3.5%
Matched Panel (median)	374,877	757.0	202,000	360.0	38,829	76.0	35.0
All (median)	41,150,001	754.0	184,000	360.0	40,474	75.0	34.0
Difference	*0.9% of total*	0.4%	9.8%	0.0%	-4.1%	0.0%	2.9%



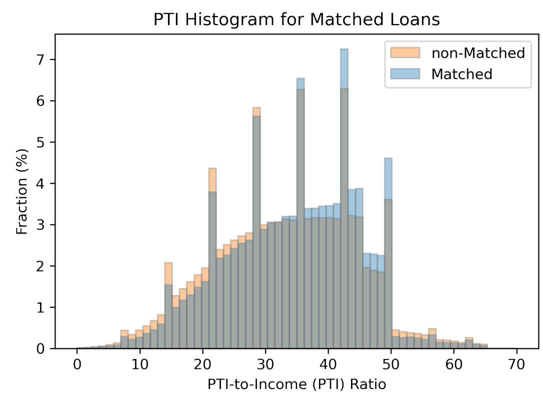
(a) Credit Score



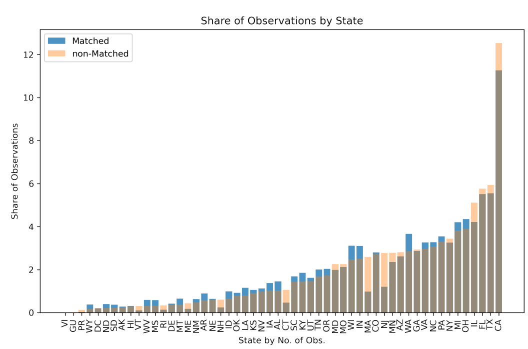
(b) Income



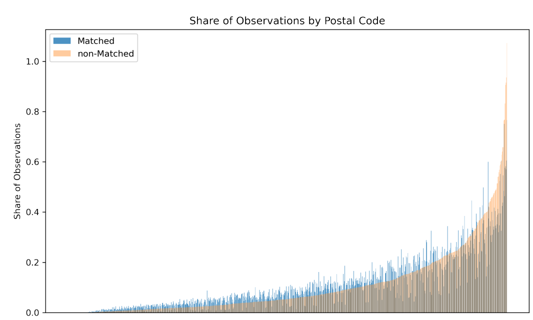
(c) Loan-to-Value (LTV)



(d) Payment-to-Income (PTI)



(e) Share of Observations by State



(f) Share of Observations by ZIP code

Figure A.2: Matched and Non-Matched Data Comparison by Distribution of Key Variables

A.3 Predicted Monthly Refinancing Probabilities

Table A.10: Predicted Monthly Refinancing Probability by Income Quintile (Simple Rate Gap)

Dependent	Refinancing Indicator × 100			
Equation	(1)	(2)	(3)	(4)
Income Quintile 1	1.923[a]	1.830[a]	1.843[a]	2.000[a]
	(0.064)	(0.047)	(0.047)	(0.055)
Income Quintile 2	2.135[a]	2.146[a]	2.167[a]	2.297[a]
	(0.044)	(0.039)	(0.038)	(0.030)
Income Quintile 3	2.404[a]	2.477[a]	2.502[a]	2.483[a]
	(0.074)	(0.054)	(0.054)	(0.055)
Income Quintile 4	2.769[a]	2.844[a]	2.870[a]	2.644[a]
	(0.058)	(0.055)	(0.058)	(0.058)
Income Quintile 5	3.375[a]	3.403[a]	3.462[a]	3.096[a]
	(0.092)	(0.077)	(0.079)	(0.078)
Control variables				
LTV, credit, loan age, rate inc., ZIP		x	x	x
Credit tightness, HARP			x	x
Loan balance, Est. LTV				x
Observations	7,698,470	7,692,279	7,680,509	7,680,509

Notes: Superscript a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis. This table reports predicted monthly refinancing probabilities by income quintile from the LPM estimates in the main text. The sample is restricted to in-the-money observations using the simple rate-gap definition. The dependent variable is scaled by 100, so entries are in percentage points. Standard errors are clustered by ZIP code and original vintage year.

Table A.11: Predicted Monthly Refinancing Probability by Income Quintile (ADL-Adjusted Rate Gap)

Dependent	Refinancing Indicator × 100			
Equation	(1)	(2)	(3)	(4)
Income Quintile 1	1.941[a]	1.882[a]	1.948[a]	2.129[a]
	(0.073)	(0.075)	(0.067)	(0.057)
Income Quintile 2	2.459[a]	2.444[a]	2.486[a]	2.670[a]
	(0.062)	(0.041)	(0.039)	(0.042)
Income Quintile 3	2.864[a]	2.913[a]	2.916[a]	2.942[a]
	(0.062)	(0.037)	(0.036)	(0.036)
Income Quintile 4	3.262[a]	3.354[a]	3.312[a]	3.124[a]
	(0.082)	(0.057)	(0.055)	(0.052)
Income Quintile 5	3.813[a]	3.792[a]	3.761[a]	3.440[a]
	(0.097)	(0.075)	(0.074)	(0.071)
Control variables				
LTV, credit, loan age, rate inc., ZIP		x	x	x
Credit tightness, HARP			x	x
Loan balance, Est. LTV				x
Observations	1,343,894	1,342,423	1,342,278	1,342,278

Notes: Superscript a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis. This table reports predicted monthly refinancing probabilities by income quintile from the LPM estimates in the main text. The sample is restricted to in-the-money observations using the ADL-adjusted rate-gap definition. The dependent variable is scaled by 100, so entries are in percentage points. Standard errors are clustered by ZIP code and original vintage year.

A.4 Baseline LPM with Time Fixed Effects

Table A.12: Baseline LPM with Reporting-Time Fixed Effects

Dependent	Refinancing Indicator × 100					
ITM Definition	Simple Rate Gap			ADL Gap		
Time FE	Month	Quarter	Year	Month	Quarter	Year
Income Quintile 2	0.52^{a}	0.52^{a}	0.56^{a}	0.93^{a}	0.91^{a}	0.91^{a}
	(0.08)	(0.08)	(0.08)	(0.14)	(0.16)	(0.15)
Income Quintile 3	0.73^{a}	0.74^{a}	0.74^{a}	1.28^{a}	1.26^{a}	1.22^{a}
	(0.09)	(0.11)	(0.13)	(0.17)	(0.19)	(0.26)
Income Quintile 4	0.97^{a}	0.98^{a}	0.98^{a}	1.56^{a}	1.54^{a}	1.47^{a}
	(0.11)	(0.13)	(0.15)	(0.20)	(0.25)	(0.32)
Income Quintile 5	1.48^{a}	1.50^{a}	1.50^{a}	2.02^{a}	2.01^{a}	1.96^{a}
	(0.13)	(0.16)	(0.20)	(0.27)	(0.34)	(0.45)
Control variables	x	x	x	x	x	x
ZIP-code FE	x	x	x	x	x	x
Reporting-time FE	x	x	x	x	x	x
Observations	4,498,353	4,498,355	4,498,355	743,547	743,554	743,554
R-squared	0.072	0.064	0.034	0.047	0.039	0.029

Notes: Superscript a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis. This table re-estimates the full-control monthly refinancing LPM after replacing origination vintage fixed effects with reporting-time fixed effects. Columns 1–3 use the simple rate-gap in-the-money sample; columns 4–6 use the ADL-adjusted in-the-money sample. Month, quarter, and year refer to fixed effects based on the loan reporting period. All specifications include the full set of borrower and loan controls from the benchmark specification, ZIP-code fixed effects, and time-varying lender-side controls. Standard errors are clustered by ZIP code and the corresponding reporting-time fixed effect.

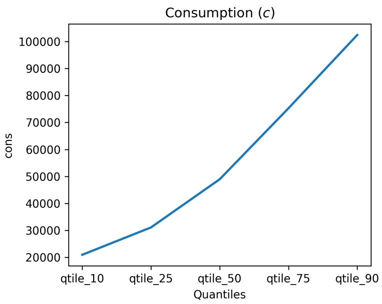
A.5 Twelve-Month Refinancing Probability Estimates

Table A.13: Twelve-Month Refinancing Probability: Regression Estimates

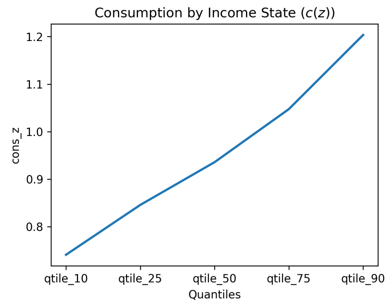
Dependent	Refinancing within 12 Months × 100	
ITM Definition	Simple Rate Gap	ADL Gap
Income Quintile 2	0.97^{c}	5.53^{a}
	(0.54)	(1.07)
Income Quintile 3	2.10^{a}	7.31^{a}
	(0.60)	(1.09)
Income Quintile 4	4.01^{a}	9.24^{a}
	(0.81)	(1.37)
Income Quintile 5	7.65^{a}	13.97^{a}
	(1.06)	(1.61)
Observations	326,147	81,467
R-squared	0.219	0.168

Notes: Superscript a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis. This table reports coefficients on income-quintile indicators relative to the bottom income quintile. The dependent variable equals 100 if the loan refinances within twelve months after first becoming in-the-money and zero otherwise. The specification includes the full set of borrower and loan controls, ZIP-code fixed effects, and origination vintage year-quarter fixed effects. Standard errors are clustered by ZIP code and original vintage year.

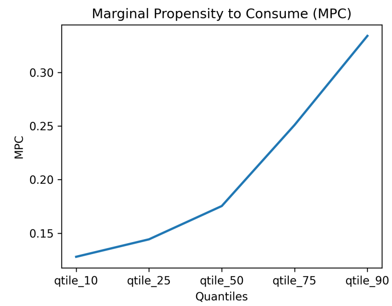
A.6 Statistic Moments of the Baseline Model



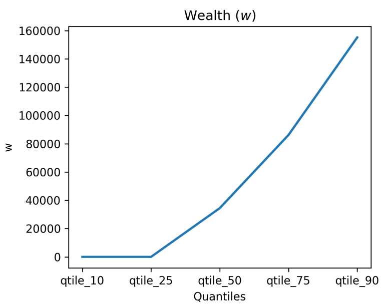
(a) Consumption



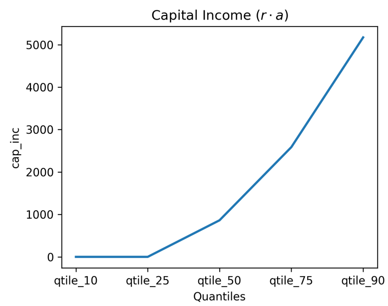
(b) Consumption to Income



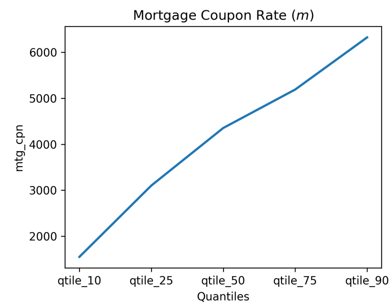
(c) MPC



(d) Wealth



(e) Capital Income



(f) Mortgage Coupon Payment

Figure A.3: Cross-Sectional Moments of the Baseline Model

B Appendix: Identification of In-the-money Loans

As described in the text, an in-the-money loan refers to a situation where the difference between the old and new interest rates after refinancing exceeds a certain threshold. This threshold represents the fixed costs associated with refinancing and option value.

$$\text{In-the-money } \textit{if } \underbrace{(\text{Interest Rate}_i^{old} - \text{Potential Refi Rate}_{i,t}^{new}) - \text{Threshold}_{i,t}}_{\equiv \text{ ADL-adjusted rate gap}} > 0$$

(i and t denote loan and time respectively)

Agarwal et al. (2013) has proposed the following tractable closed-form solution for the threshold value.

$$\begin{aligned} Threshold &= \frac{1}{\psi}\left[\phi + W\left(-\exp\left(-\phi\right)\right)\right] \\ \psi &= \sqrt{\frac{2(\rho+\lambda)}{\sigma}} \\ \phi &= 1 + \psi(\rho+\lambda)\frac{\kappa(M)/M}{(1-\tau)} \end{aligned}$$

Here, $W(\cdot)$ is the Lambert W-function, ρ is the real discount rate, λ is the expected real rate of exogenous mortgage repayment, σ is the standard deviation of the mortgage rate, $\kappa(M)/M$ is the ratio of the tax-adjusted refinancing cost and the remaining mortgage value, and τ is the marginal tax rate. The values of the parameters suggested by Agarwal et al., 2013 are $\sigma = 0.0109$, $\tau = 0.28$, $\rho = 0.05$.

$\kappa(M)$ is the cost entailed in refinancing, which is defined by:

$$\kappa(M) = F + fM\left[1 - \frac{\tau}{\theta+\rho+\pi}\left\{\frac{1-\exp\left(-(\theta+\rho+\pi)N\right)}{N}\frac{\rho+\pi}{\theta+\rho+\pi} + \theta\right\}\right]$$

F is the fixed cost of refinance, f is the points divided by 100, τ is the marginal tax rate, θ is the expected arrival rate of a full deductibility event, such as moving and refinancing. Using the parameter suggested by the paper $F = 2000, f = 0.01, \theta = \mu + 0.1 = 0.2$, the cost can be approximated as follows.

$$\kappa(M) = 2000 + 0.007905M$$

Lastly, Lambda is defined as the expected rate of decline in the real principal of the mortgage for reasons other than rate-reducing refinancing, and is defined as:

$$\lambda = \mu + \frac{i^{old}}{\exp(i^{old}\Gamma) - 1} + \pi,$$

Γ is the remaining life of the existing mortgage in years, i^{old} is the current (old) mortgage interest rate for existing loan, μ is the hazard of relocation, and π is the inflation rate. The parameter values used are $\mu = 0.1$ and $\pi = 0.03$, following the paper.

C Appendix: Robustness Checks

C.1 Credit Score and PTI Bins

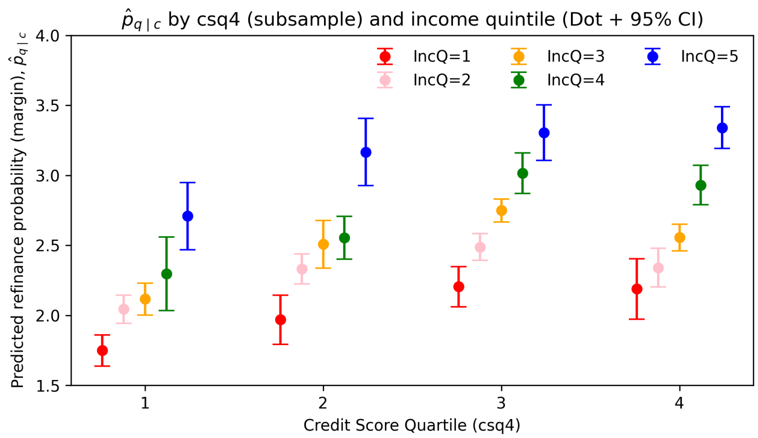


Figure C.4: Refinancing Propensity by Income Quintile and Leverage Ratio

Notes: The sample is split into quartiles of the borrower credit score. Within each credit-score quartile, I re-estimate the baseline linear probability model using the same set of controls and fixed effects as in the main specification. The figure reports the estimated refinancing probabilities for each income quintile within a given credit-score bin. Dots represent point estimates and bars denote 95

Table C.14: Refinancing Probability by Income Quintile within PTI Bins

PTI bin	< 36	$[36, 43)$	$[43, 45)$	$[45, 50)$	$[50, 60)$	≥ 60
Income Q2	0.388**	0.166	0.160	0.451**	0.466**	1.168**
	(0.109)	(0.105)	(0.182)	(0.131)	(0.143)	(0.308)
Income Q3	0.579**	0.370*	0.240	0.794**	1.028**	1.536*
	(0.122)	(0.164)	(0.249)	(0.187)	(0.229)	(0.708)
Income Q4	0.970**	0.378	0.095	0.930**	1.398**	1.396*
	(0.119)	(0.253)	(0.323)	(0.279)	(0.277)	(0.678)
Income Q5	1.409**	0.973**	1.549*	1.593**	1.214*	7.174**
	(0.153)	(0.284)	(0.640)	(0.360)	(0.557)	(1.521)

Notes: The sample is partitioned by payment-to-income (PTI) bins corresponding to common underwriting thresholds: PTI < 36, $[36, 43)$, $[43, 45)$, $[45, 50)$, and $[50, 60)$, and ≥ 60. Within each PTI bin, I re-estimate the baseline linear probability model using the same set of controls and fixed effects as in the baseline specification. Reported coefficients correspond to income-quintile indicators relative to the lowest income quintile (Q1). Standard errors are reported in parentheses and significance levels are indicated by $*\,p < 0.10$, $**\,p < 0.05$, and $***\,p < 0.01$.

C.2 Origination Lender-Year Fixed Effects

This section uses the Freddie Mac-HMDA linked panel to examine whether the refinancing gradient reflects sorting into different origination lenders or different lender-year vintages. The HMDA match identifies the lender that originated the Freddie Mac loan, which allows me to compare borrowers within the same origination lender-year cell while also controlling for the origination quarter and borrower location.

Table C.15: HMDA-Freddie Mac Lender Match Coverage

Statistic	Count
Origination cohorts covered	2000–2022
Monthly loan observations	12,473,999
Monthly observations with origination lender ID	2,658,950
Unique loans	374,877
Unique loans with origination lender ID	90,637
Unique origination lenders	6,828
Rate-gap ITM observations with origination lender ID	1,781,857
ADL-gap ITM observations with origination lender ID	342,274

Notes: The table summarizes the HMDA-Freddie Mac linked panel used for the origination-lender robustness checks. Monthly observations are loan-month observations. ITM denotes the in-the-money samples used in the simple rate-gap and ADL-gap linear probability models.

The estimating equation is

$$Refi_{it} = \sum_{q=2}^{5} \beta_q IncomeQuintile_{iq} + X_{it}'\Gamma + \alpha_{z(i)} + \lambda_{v(i)} + \mu_{\ell(i)\times y(i)} + \varepsilon_{it}, \tag{21}$$

where $Refi_{it}$ is the monthly refinancing indicator multiplied by 100, $IncomeQuintile_{iq}$ denotes income-quintile indicators relative to the lowest quintile, and X_{it} contains the baseline loan and borrower controls. The fixed effects include ZIP-code fixed effects $\alpha_{z(i)}$, origination year-quarter fixed effects $\lambda_{v(i)}$, and origination lender-year fixed effects $\mu_{\ell(i)\times y(i)}$, defined by HMDA origination lender ID interacted with origination year. Standard errors are clustered by ZIP code and origination year-quarter, as in the HMDA regression output.

Table C.16 and Table C.17 show that the income gradient remains after adding origination lender-year fixed effects. In the full-control specification, borrowers in the highest income quintile are 1.255 percentage points more likely to refinance than borrowers in the lowest quintile under the simple rate-gap definition, and 2.581 percentage points more likely under the ADL-gap definition. The coefficients for income quintiles 2 through 5 are positive and statistically significant in both tables.

These estimates address the concern that the benchmark gradient is driven by low-income borrowers being originated by lenders or lender-year vintages with systematically lower later refinancing activity. For example, the results are not explained solely by low-income borrowers initially receiving loans from lenders with different business models, servicing technologies, borrower pools, or origination pricing environments. Because the gradient remains within the same origination lender-year cells, initial lender composition and lender-vintage-specific underwriting conditions do not appear to mechanically generate the main result. The remaining gradient is therefore consistent with borrower-side refinancing frictions. This exercise does not directly rule out refinance-stage lender outreach, screening, or approval frictions, but it narrows the lender-side explanation to channels operating after origination rather than to sorting into original lenders.

Table C.16: Average Refinancing Rate by Income Quintile with Origination Lender-Year Fixed Effects (Simple Rate Gap)

Dependent	Refinancing Indicator × 100			
Specification	(1)	(2)	(3)	(4)
Income Quintile 2	0.315[a]	0.432[a]	0.493[a]	0.392[a]
	(0.092)	(0.093)	(0.112)	(0.122)
Income Quintile 3	0.636[a]	0.841[a]	0.851[a]	0.465[a]
	(0.143)	(0.132)	(0.140)	(0.164)
Income Quintile 4	1.075[a]	1.325[a]	1.425[a]	0.778[a]
	(0.109)	(0.123)	(0.123)	(0.156)
Income Quintile 5	1.649[a]	1.867[a]	2.076[a]	1.255[a]
	(0.114)	(0.134)	(0.148)	(0.183)
Control variables and fixed effects				
Origination year-quarter FE	x	x	x	x
Origination lender-year FE	x	x	x	x
ZIP FE		x	x	x
LTV, credit, loan age, rate inc.		x	x	x
Credit tightness, HARP			x	x
Loan balance, Est. LTV				x
Observations	1,781,560	1,780,614	1,778,383	1,778,383
Adj. R-squared	0.020	0.031	0.032	0.033

Notes: Superscript a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis. Standard errors are clustered by zipcode and origination year-quarter. The sample is restricted to loan-month observations with a positive simple rate gap and non-missing origination lender-year fixed effect. Origination lender-year fixed effects are defined by HMDA origination lender ID interacted with origination year. Specification (4) corresponds to the full-control specification.

Table C.17: Average Refinancing Rate by Income Quintile with Origination Lender-Year Fixed Effects (ADL Gap)

Dependent	Refinancing Indicator × 100			
Specification	(1)	(2)	(3)	(4)
Income Quintile 2	0.755[a]	1.100[a]	1.111[a]	1.057[a]
	(0.139)	(0.176)	(0.226)	(0.226)
Income Quintile 3	1.308[a]	1.887[a]	1.841[a]	1.442[a]
	(0.172)	(0.198)	(0.277)	(0.302)
Income Quintile 4	1.841[a]	2.708[a]	2.775[a]	2.054[a]
	(0.196)	(0.244)	(0.272)	(0.335)
Income Quintile 5	2.448[a]	3.228[a]	3.506[a]	2.581[a]
	(0.211)	(0.238)	(0.314)	(0.395)
Control variables and fixed effects				
Origination year-quarter FE	x	x	x	x
Origination lender-year FE	x	x	x	x
ZIP FE		x	x	x
LTV, credit, loan age, rate inc.		x	x	x
Credit tightness, HARP			x	x
Loan balance, Est. LTV				x
Observations	342,201	341,873	341,801	341,801
Adj. R-squared	0.026	0.031	0.032	0.033

Notes: Superscript a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis. Standard errors are clustered by zipcode and origination year-quarter. The sample is restricted to loan-month observations classified as in the money under the ADL rule and with non-missing origination lender-year fixed effect. Origination lender-year fixed effects are defined by HMDA origination lender ID interacted with origination year. Specification (4) corresponds to the full-control specification.

C.3 Changes in Income Quintile

Table C.18: Income Quintile Transition Matrix

Data	Income Quintile	To 1	To 2	To 3	To 4	To 5	Total (%)
PSID[1]	From 1	68.9	14.8	6.8	4.0	5.6	100.0
	2	26.2	35.7	22.6	9.6	6.0	100.0
	3	12.6	17.8	33.2	24.9	11.5	100.0
	4	4.7	7.6	17.2	38.6	31.9	100.0
	5	2.4	1.6	3.2	10.0	82.7	100.0
Loan Panel[2]	From 1	43.6	24.6	15.0	10.0	6.8	100.0
	2	20.9	30.4	22.7	15.4	10.7	100.0
	3	9.3	22.4	28.9	23.1	16.3	100.0
	4	4.9	11.4	22.8	33.4	27.5	100.0
	5	2.9	6.1	13.0	25.6	52.4	100.0

Notes: (1) Panel Study of Income Dynamics. Changes of mortgagor's income over two years, 2005–2021. (2) Changes from origination to refinancing.

Table C.19: Using New Loan's Income Quintile

Dependent Equation	Refinancing Indicator × 100 (1)	(2)	(3)	(4)	(5)
Income Quintile 2	0.261^a	0.340^a	0.336^a	0.286^a	0.309^a
	(0.052)	(0.068)	(0.072)	(0.075)	(0.077)
Income Quintile 3	0.563^a	0.723^a	0.731^a	0.573^a	0.610^a
	(0.049)	(0.066)	(0.065)	(0.074)	(0.073)
Income Quintile 4	0.858^a	1.090^a	1.104^a	0.860^a	0.895^a
	(0.051)	(0.057)	(0.057)	(0.089)	(0.089)
Income Quintile 5	1.191^a	1.504^a	1.522^a	1.211^a	1.252^a
	(0.073)	(0.086)	(0.088)	(0.138)	(0.137)
Control variables					
LTV, credit, loan age, rate inc., ZIP		x	x	x	x
Credit tightness			x	x	x
HARP, loan balance				x	x
Est. LTV					x
Observations	3,691,616	3,691,312	3,687,034	3,687,034	3,687,034
R-squared	0.015	0.024	0.024	0.024	0.025

Notes: Superscript a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis. This table reports LPM coefficients on income-quintile indicators relative to the bottom income quintile. Income quintiles are assigned using the borrower's income quintile on the new refinance loan. The sample is restricted to in-the-money observations under the simple rate-gap definition. Standard errors are clustered by ZIP code and original vintage year.

Table C.20: Restricting Samples Without Income Quintile Changes

Dependent Equation	Refinancing Indicator × 100 (1)	 (2)	 (3)	 (4)	 (5)
Income Quintile 2	0.548[a]	0.778[a]	0.820[a]	0.794[a]	0.802[a]
	(0.078)	(0.104)	(0.111)	(0.110)	(0.111)
Income Quintile 3	0.862[a]	1.226[a]	1.281[a]	1.114[a]	1.117[a]
	(0.112)	(0.145)	(0.140)	(0.156)	(0.152)
Income Quintile 4	1.412[a]	1.973[a]	2.013[a]	1.746[a]	1.659[a]
	(0.107)	(0.133)	(0.133)	(0.166)	(0.165)
Income Quintile 5	2.111[a]	2.740[a]	2.761[a]	2.460[a]	2.311[a]
	(0.140)	(0.162)	(0.167)	(0.218)	(0.216)
Control variables					
LTV, credit, loan age, rate inc., ZIP		x	x	x	x
Credit tightness, HARP			x	x	x
Loan balance				x	x
Est. LTV					x
Observations	1,334,728	1,333,447	1,331,993	1,331,993	1,331,993
R-squared	0.019	0.030	0.029	0.029	0.030

Notes: Superscript a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis. This table reports LPM coefficients on income-quintile indicators relative to the bottom income quintile. The sample is restricted to loans for which the original-loan income quintile equals the new-loan income quintile. The sample is restricted to in-the-money observations under the simple rate-gap definition. Standard errors are clustered by ZIP code and original vintage year.

Table C.21: Five-Year Rolling Window Estimates

Rolling window end year	2005	2007	2009	2011	2013	2015	2017	2019	2021
Income Quintile 2	0.715[a]	0.373[a]	0.449[a]	0.434[a]	0.435[a]	0.396[a]	0.357[c]	0.248	-0.155
	(0.127)	(0.082)	(0.052)	(0.051)	(0.063)	(0.091)	(0.206)	(0.186)	(0.268)
Income Quintile 3	0.853[a]	0.585[a]	0.762[a]	0.848[a]	0.793[a]	0.483[a]	0.384[b]	0.184	-0.303
	(0.150)	(0.072)	(0.067)	(0.061)	(0.081)	(0.108)	(0.180)	(0.279)	(0.319)
Income Quintile 4	1.071[a]	0.729[a]	0.996[a]	1.104[a]	1.074[a]	0.591[a]	0.632[a]	0.115	-0.381
	(0.203)	(0.121)	(0.124)	(0.099)	(0.125)	(0.117)	(0.214)	(0.330)	(0.385)
Income Quintile 5	1.674[a]	1.008[a]	1.307[a]	1.635[a]	1.594[a]	0.934[a]	0.888[a]	0.290	-0.304
	(0.214)	(0.108)	(0.139)	(0.117)	(0.153)	(0.177)	(0.286)	(0.400)	(0.487)

Notes: Superscript a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis. Each column reports the full-control LPM estimated on a five-year rolling window ending in the indicated year. The dependent variable is the monthly refinancing indicator scaled by 100, and the sample is restricted to in-the-money observations under the simple rate-gap definition. The specification includes the full set of borrower and loan controls, ZIP-code fixed effects, and origination vintage year-quarter fixed effects. Standard errors are clustered by ZIP code and original vintage year.

C.4 Survey of Consumer Finances (SCF)

Finally, I utilized the Survey of Consumer Finances (SCF) to examine whether the same income gradient appears in survey data. While the SCF does not provide panel data and is conducted as a triennial survey, it allows us to observe the year of the most recent refinancing for currently held mortgages. This enables us to derive annual refinancing rates.

Figure C.6 Panel (a) shows the refinancing rates by income quintile, calculated using pooled data from seven surveys conducted between 2004 and 2022. When compared to the Freddie Mac data, the SCF data similarly reveals that lower-income households exhibit a lower propensity to refinance. However, the SCF data overall shows higher refinancing rates, likely due to differences in data coverage, as will be discussed later.

Conversely, Panel (b) displays refinancing rates divided into quintiles based on the leverage (loan balance to income) ratio as per Chen et al. (2020). Interestingly, this presents a contrasting pattern: the SCF data indicates higher refinancing activity as the leverage ratio increases, whereas the Freddie Mac data shows a decrease in refinancing propensity.

The divergence in these results can be attributed to two main factors. First, the nature of the datasets differs significantly. The SCF is a survey-based triennial dataset that uses current household income, whereas the Freddie Mac data is an administrative panel dataset based on the borrower's income at the time of loan origination. Consequently, differences in the calculation of the loan-to-income ratio, due to variations between household and individual income and current versus past income, complicate direct comparisons.

More importantly, the scope of households covered by each dataset appears to contribute to these contrasting results. As shown in Figure C.7, the types of mortgages in the U.S. are broadly categorized into conventional and non-conventional mortgages. Conventional mortgages are further divided into conforming and jumbo (non-conforming) loans. Conforming loans meet the guidelines set by the Federal Housing Finance Agency (FHFA), including loan limits and credit requirements, whereas jumbo loans exceed these limits. Conforming loans are mostly acquired in the secondary market by government-sponsored enterprises (GSEs) like Fannie Mae and Freddie

Table C.22: Regression by Loan Age Restrictions

Dependent	Refinancing Indicator x 100			
Loan Age ≤	1 yr	2 yrs	3 yrs	5 yrs
IncomeQuintile 2	0.561[a]	1.220[a]	1.229[a]	0.962[a]
	(0.245)	(0.146)	(0.0951)	(0.0732)
IncomeQuintile 3	1.235[a]	1.963[a]	2.041[a]	1.741[a]
	(0.250)	(0.153)	(0.102)	(0.0783)
IncomeQuintile 4	2.192[a]	2.875[a]	2.752[a]	2.403[a]
	(0.259)	(0.161)	(0.111)	(0.0854)
IncomeQuintile 5	2.912[a]	3.642[a]	3.654[a]	3.283[a]
	(0.292)	(0.180)	(0.127)	(0.0981)
Constant	0.276	-0.373	-0.244	-0.0230
	(0.589)	(0.354)	(0.263)	(0.200)
Observations	97,515	374,996	670,264	945,099
R-squared	0.022	0.015	0.014	0.012

Notes: a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis. The results base on the benchmark regression (5); control variables are LTV, credit score, rate gap, credit tightness, HARP, loan balance to income, estimated LTV.

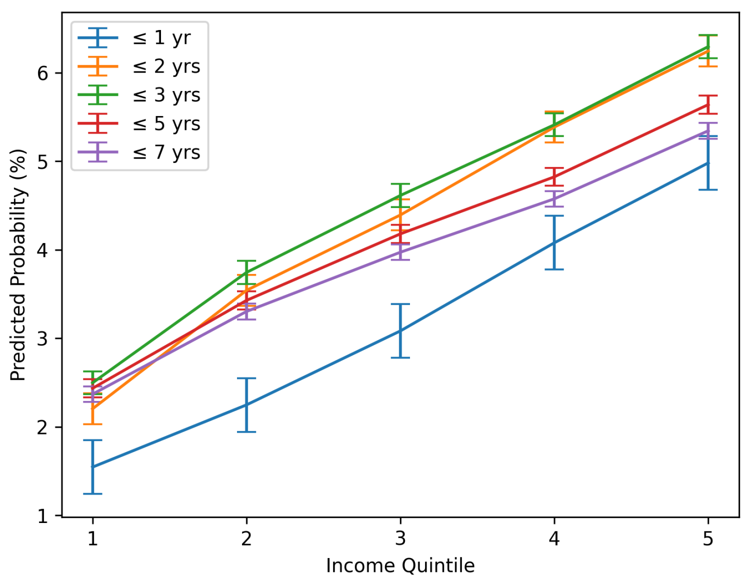
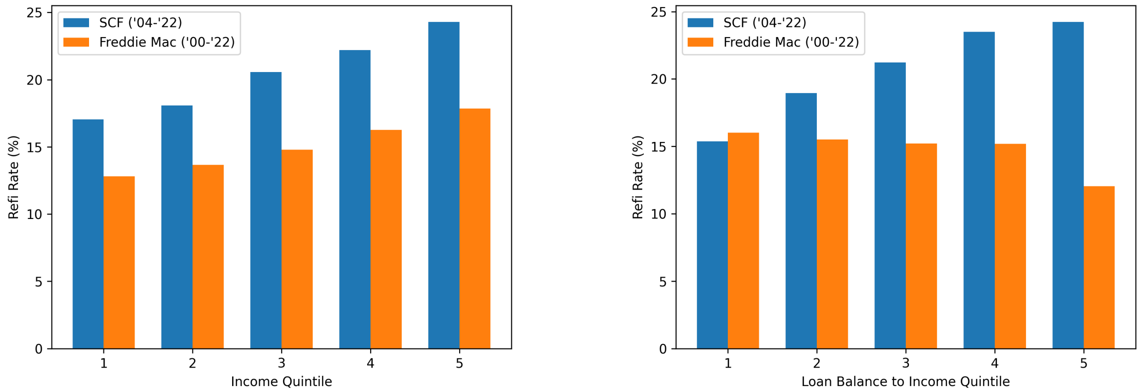


Figure C.5: Predicted Refinancing Probability by Loan Age

Notes: This plot shows the predicted monthly refinancing probability by loan age, using Equation 2. The results are based on the benchmark regression (5); control variables are LTV, credit score, rate gap, credit tightness, HARP, loan balance to income, and estimated LTV. The 95% confidence intervals are represented by error bars.



(a) Refinancing Rate by Income (b) Refinancing Rate by Leverage Ratio

Figure C.6: Refinancing Propensity by Income Quintile and Leverage Ratio

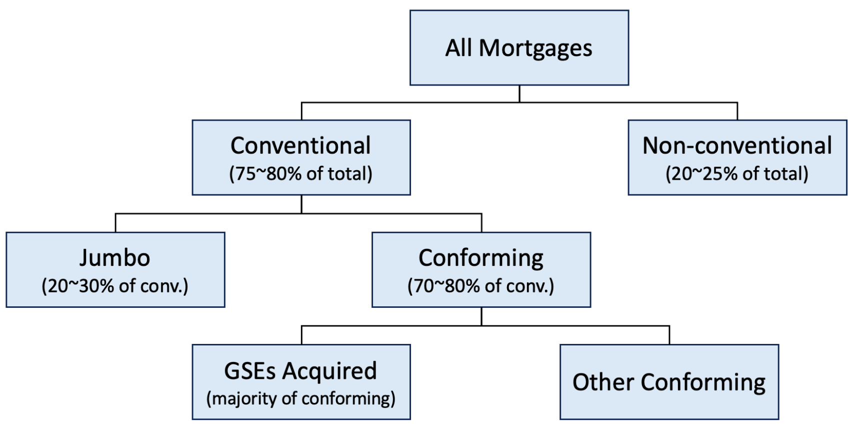


Figure C.7: Mortgage Types in the U.S.

Notes: Market shares of each mortgage type vary over time, and the numbers provided are approximate, referencing the National Mortgage Database (NMDB), web searches.

Mac. The Freddie Mac data that I used primarily represents about 70% of conventional loans, accounting for approximately 60% of all loans.

However, the loans not covered by the Freddie Mac data, particularly the jumbo loans, which constitute about 30% of conventional loans, seem to be influencing the results. Jumbo loans, exceeding conforming loan limits, are predominantly held by high-income, wealthy individuals. Since the SCF samples all households, including jumbo loan holders, it exhibits significantly higher income levels, even considering household versus borrower income as shown in Table C.23.

Table C.23: Freddie Mac (Borrower) and SCF (Household) Income by Leverage Ratio

Loan-to-income Quintile	Median Income ($)		Mean Income ($)	
	Freddie Mac	SCF	Freddie Mac	SCF
1	53,143	356,204	68,713	1,790,221
2	41,095	161,175	47,831	366,883
3	35,438	119,000	40,494	189,615
4	31,529	95,000	35,493	142,475
5	28,984	58,396	31,971	83,983

Notes: The SCF uses current household income, while Freddie Mac uses the borrower's individual income at the time of loan origination. The statistics are based on pooled SCF 2004–2022 triennial public data and Freddie Mac loans originated from 2000 to 2022.

Thus, assuming comparability despite the inherent differences in data characteristics, the sample including all households shows increased refinancing with higher leverage. In contrast, narrowing the analysis to middle and lower-income households reveals an opposite and intriguing pattern. Highly leveraged households might have a high demand for cash-out refinancing due to idiosyncratic shocks, but those with higher absolute incomes might experience fewer frictions related to low income, such as liquidity constraints for closing costs or financial illiteracy, which affect refinancing decisions.

Regarding policy relevance, which dataset is more relevant? If the limitations of survey data is disregard, the SCF offers a more comprehensive coverage. However, if monetary policy focuses on relatively liquidity-constrained, high-MPC middle and lower-income households, the subset I used would be more significant for policymakers. Specifically, frictions unique to lower-income households can disrupt the transmission channels of monetary policy, making the refinancing behavior of the wealthy less pertinent. Comparing refinancing propensity using consistent income standards across different data scopes could be an interesting area for further research.

C.5 Estimating Unrealized Savings by Constructing Counterfactual Rate Gaps

As an alternative approach to quantifying unrealized savings, I estimate income-group-specific rate gaps directly from the population of in-the-money loan-months, rather than relying on the matched-loan sample. This method constructs a counterfactual refinancing rate for each borrower and translates it into an implied monthly payment reduction.

Step 1: Estimating the predicted rate gap by income group. I regress the rate gap $RateGap_{it} \equiv c_{it} - m_{it}$ on income quintile indicators and the full set of controls, restricting the sample to in-the-money observations ($gap > 0$):

$$RateGap_{it} = \alpha + \sum_{q=2}^{5} \hat{\beta}_q \cdot \mathbf{1}\{IncomeQ_i = q\} + \Gamma' \cdot X_{it} + \nu_{z(i)} + \lambda_{v(i)} + \epsilon_{it} \tag{22}$$

where c_{it} is borrower i's contracted coupon rate and m_{it} is the prevailing obtainable rate. The coefficients $\hat{\beta}_q$ capture the income-quintile gradient in the rate gap after controlling for borrower and loan characteristics, ZIP fixed effects, and vintage fixed effects.

Step 2: Constructing the counterfactual refinancing rate. Using the estimated income-group predicted rate gap $\widehat{RateGap}_{q(i)}$, I construct a counterfactual new coupon rate for each in-the-money loan-month:

$$\hat{m}_{it}^{new} = c_{it} - \widehat{RateGap}_{q(i)} \tag{23}$$

This counterfactual rate represents what a borrower in income quintile q would pay if they refinanced and captured the income-group-specific rate benefit. I then translate this into an implied monthly payment reduction using loan-specific balance B_{it}, current coupon c_{it}, and remaining term N_{it}:

$$\widehat{\Delta Pay}_{it} = PMT(B_{it}, c_{it}, N_{it}) - PMT\left(B_{it}, \hat{m}_{it}^{new}, N_{it}\right) \tag{24}$$

Step 3: Aggregating unrealized savings by income group. I aggregate $\widehat{\Delta Pay}_{it}$ across in-the-money loan-months within each income quintile and normalize by average income to obtain the payment reduction as a share of income.

The results are reported in Table C.24. Consistent with the matched-loan estimates, the implied monthly payment reduction is largest relative to income for the lowest-income borrowers, amounting to approximately 7.6% of monthly income for the bottom quintile compared with 4.5% for the top quintile. The similarity of these estimates to those from the matched-loan approach corroborates the robustness of the main finding: foregone refinancing savings are disproportionately large relative to income for lower-income households, precisely those with the highest marginal propensity to consume.

Income Quintile	Avg. Income ($)	Avg. Balance ($)	Est. Rate Gap (pp)	Mthly. Pymt. Reduction ($)	Reduction / Income (%)
1	16,579	89,989	1.01	100.14	7.61
2	28,121	145,956	1.00	158.26	6.58
3	39,391	195,149	0.99	207.84	6.12
4	54,772	253,904	0.97	268.55	5.56
5	98,047	312,712	0.96	379.16	4.49

Payment savings assume refinancing resets the loan to a 30-year FRM term. Rate gaps are estimated using Eq. (5) (full controls). Freddie Mac statistics suggest that about 70–75% of refinances keep the term length unchanged. Sample is restricted to in-the-money observations ($gap > 0$).

Table C.24: Unrealized Savings under the Counterfactual Rate-Gap Approach

D Appendix: Additional Results on Differential Refinancing Response to Monetary Policy Shocks

This appendix reports additional results for the monetary-policy refinancing exercise in **??**. The purpose is twofold. First, I verify that the high-frequency monetary-policy shocks generate a strong first-stage movement in mortgage rates and a sizable average refinancing response. Second, I show that the short-run income gradient documented in the main text is robust to alternative refinancing-response measures and to a coarser high-versus-low income grouping.

First stage and average refinancing response. I first report the first-stage relationship between high-frequency monetary-policy shocks and the 30-year fixed mortgage rate. Table D.25 shows that the Swanson high-frequency factors are strongly related to movements in mortgage rates. The first-stage F-statistic is well above conventional weak-instrument thresholds, indicating that the monetary surprises have substantial explanatory power for mortgage-rate changes. The signs are also economically consistent: shocks associated with tighter monetary policy raise mortgage rates, while the transformed easing shock used in the local projections is normalized so that a positive value corresponds to a decline in mortgage rates.

Table D.25: First-stage Estimates

Dependent		Δ Real Mortgage Rates of 30-year FRM	
High-frequency Factors		Survey of Professional Forecasters	
- FFR	0.0165[a]	- ΔGDP (2 yrs)	-0.0341[a]
	(0.00)		(0.00)
- Forward Guidance	0.0290[a]	- Unemp. (2 yrs)	0.0273[a]
	(0.00)		(0.00)
- LSAP	0.0344[a]	- CPI (1 yrs)	-0.0347[a]
	(0.00)		(0.00)
ΔHouse Price	0.0005[a]	- CPI (2 yrs)	-0.0832[a]
	(0.00)		(0.01)
Constant	0.0427[a]	Unemp.	0.00641[a]
	(0.02)		(0.00)
Observations	46,351	F-statistic	2,803.820
R-squared	0.358	Prob. > F	0.000

Notes: Superscript a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis.

I next estimate the average refinancing response, pooling across income groups. This exercise is not used to identify the income gradient, but it provides a useful benchmark for the magnitude of the differential response in the main text. The specification is the same LP-IV design used in **??**, except that the dependent variable is the aggregate regional refinancing response rather than the ZIP3-by-income response. The preferred dependent variable is the cumulative sum of quarterly refinancing rates,

$$\rho_{zt}^{h} = \sum_{k=0}^{h} 100 \times \frac{x_{z,t+k}}{y_{z,t+k}},$$

where x_{zt} is the number of refinances in ZIP3 region z and quarter t, and y_{zt} is the corresponding outstanding mortgage stock.

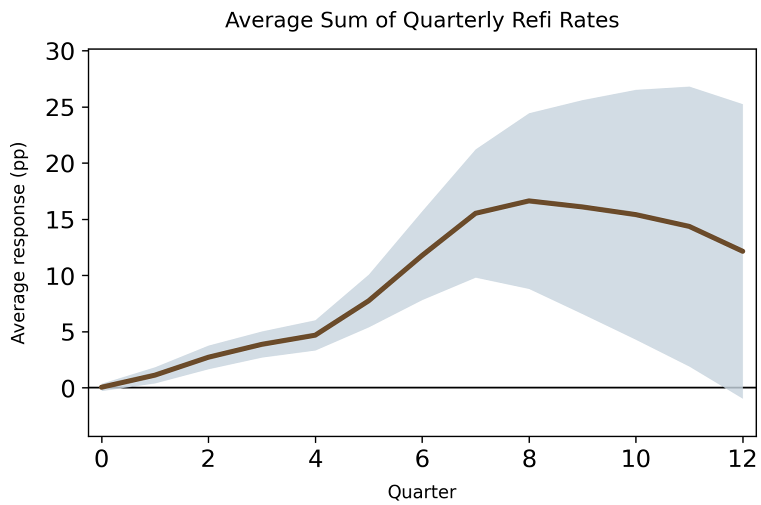


Figure D.8: Average Refinancing Response to a Monetary Policy Easing

Notes: The figure reports the average cumulative refinancing response to a 1 percentage point monetary-policy-induced decline in the 30-year mortgage rate. The dependent variable is the cumulative sum of quarterly refinancing rates. The shaded region denotes 90 percent confidence intervals based on two-way clustering by ZIP3 region and quarter.

The average refinancing response to a 1% interest rate falls. It indicates that cumulatively 17% of all mortgages are refinanced 7 quarters after the shock. After the eighth quarter, the cumulative ratio decreases because new mortgages increase the total number of outstanding loans while the number of refinancing decreases, gradually lowering the ratio below the unconditional ratio. Although the types of mortgages covered by the data and the sample periods differ from Eichenbaum et al. (2022)'s results – the response peaks at 2-10%[14] by the fifth quarter following an 50 bp mortgage-rate cut, the result is consistent with their findings.

[14]They estimated the refinancing rate depending on the rate gaps between existing loans and the market rate, which makes the estimated range larger.

Figure D.8 shows the average refinancing response to a 1 percentage point decline in the 30-year mortgage rate. The cumulative response peaks at 17.2 percentage points after seven quarters After the peak, the cumulative response declines as refinance originations fall and new mortgage originations expand the outstanding-loan stock, mechanically lowering the cumulative refinancing rate. The magnitude is broadly comparable to Eichenbaum et al. (2022), though the comparison is not one-for-one. They report cumulative refinancing responses to a 50 basis point mortgage-rate cut that peak at roughly 2–10 percent after about five quarters, depending on borrowers' initial rate gaps. Because my estimates are normalized to a 1 percentage point decline in the mortgage rate, the 17.2 percentage point peak is of a similar order of magnitude after allowing for this difference in scaling. Differences in sample coverage, mortgage types, time period, and response construction further limit a direct comparison.

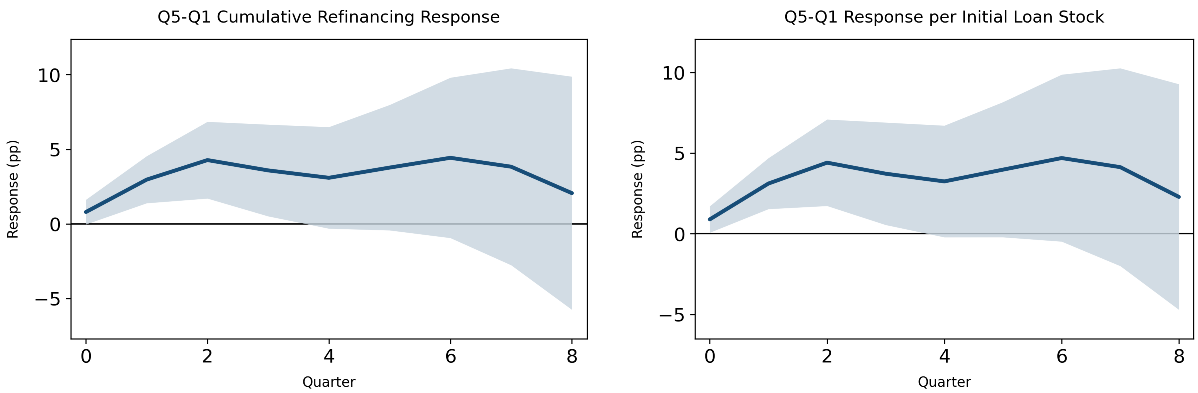
Alternative refinancing-response measures. The main text uses the cumulative sum of quarterly refinancing rates because it closely matches the refinancing-rate construction in Eichenbaum et al. (2022). I now verify that the short-run income gradient is not driven by this normalization. I consider two alternative measures. The first normalizes cumulative refinances by the average outstanding stock over the response window:

$$\rho_{zqt}^{h,1} = 100 \times \frac{\sum_{k=0}^{h} x_{zq,t+k}}{\left(\sum_{k=0}^{h} y_{zq,t+k}\right)/(h+1)}. \tag{25}$$

The second normalizes cumulative refinances by the pre-shock outstanding stock:

$$\rho_{zqt}^{h,2} = 100 \times \frac{\sum_{k=0}^{h} x_{zq,t+k}}{y_{zq,t-1}}. \tag{26}$$

The first measure allows the outstanding stock to evolve over the response horizon, while the second fixes the denominator at the pre-shock stock. Both measures are estimated using the same date fixed effects, ZIP3-by-income fixed effects, controls, instruments, weights, and two-way clustered standard errors as in the main specification.



(a) Average-stock denominator

(b) Pre-shock denominator

Figure D.9: Robustness: Q5–Q1 Differential Response under Alternative Refinancing Measures

Notes: The figure reports the Q5–Q1 differential refinancing response to a 1 percentage point monetary-policy-induced decline in the 30-year mortgage rate. Panel (a) uses cumulative refinances divided by the average outstanding stock over the response window. Panel (b) uses cumulative refinances divided by the pre-shock outstanding stock. The shaded regions denote 90 percent confidence intervals based on two-way clustering by ZIP3 region and quarter.

Figure D.9 shows that the short-run income gradient is essentially unchanged under both alternative response measures. Under the average-stock denominator, the Q5–Q1 differential response reaches 4.28 percentage points after two quarters. Under the pre-shock denominator, it reaches 4.41 percentage points. In both cases, the response is positive and statistically significant over the first several quarters, closely matching the preferred specification in the main text.

Alternative income grouping. As a final robustness check, I replace the linear income-rank specification with a coarser grouping of income cells. I define a low-income group as Q1–Q2, a middle-income group as Q3, and a high-income group as Q4–Q5. I then estimate the same LP-IV specification after replacing $\tilde{q}$ with a group-rank variable equal to zero for the low-income group, one-half for the middle group, and one for the high-income group. The coefficient is therefore interpretable as the high-minus-low differential response to the monetary-policy shock.

Figure D.10 confirms the same pattern. The high-minus-low differential response is positive on impact, rises to about 3.2 percentage points after two quarters, and remains statistically significant through horizon three. The magnitude is slightly smaller than the Q5–Q1 specification because the grouped comparison averages over the bottom two and top two quintiles, but the timing and sign

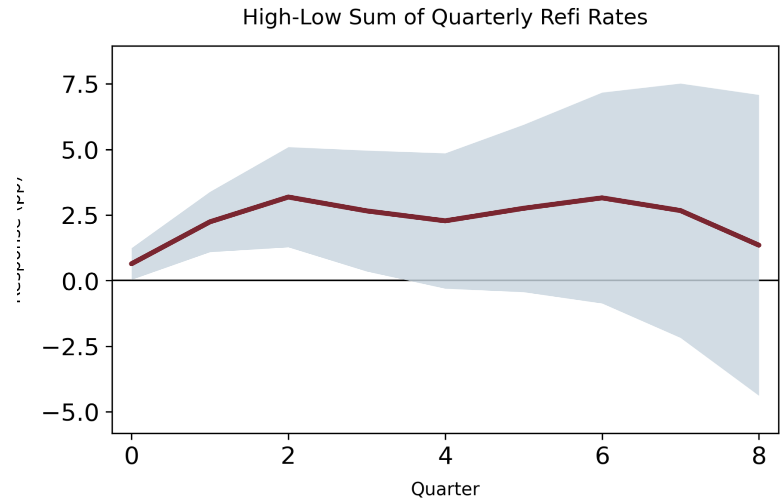


Figure D.10: Robustness: High–Low Differential Refinancing Response

Notes: The figure reports the high-minus-low differential refinancing response to a 1 percentage point monetary-policy-induced decline in the 30-year mortgage rate. The low-income group consists of Q1–Q2, the middle group consists of Q3, and the high-income group consists of Q4–Q5. The dependent variable is the cumulative sum of quarterly refinancing rates. The shaded region denotes 90 percent confidence intervals based on two-way clustering by ZIP3 region and quarter.

are nearly identical. Overall, the appendix results show that the main monetary-policy finding is robust: refinancing responses to easing shocks are concentrated among higher-income borrowers in the first year after the shock.

E Appendix: IRF Construction

This appendix describes how I construct the impulse responses reported in the quantitative section. The calculation compares two transition paths within a fixed model environment: an unshocked path and a path with a one-percentage-point decline in the policy rate at time zero. Let j index the model environment, where $j = B$ denotes the baseline calibration and $j = CF$ denotes a counterfactual environment such as the Byrne-scaled attention economy. Let $z \in \{0, 1\}$ index the interest-rate path, with $z = 0$ denoting the unshocked path and $z = 1$ denoting the rate-cut path.

The state vector is

$$S \equiv (W, F, c, s),$$

where W is liquid wealth, F is the outstanding mortgage balance, c is the coupon on the current mortgage, and s is the income state. The initial cross-sectional distribution is denoted by $g_0(S)$. In all policy experiments, the shocked and unshocked economies start from the same initial distribution g_0. The short rate in the unshocked path is r_t^0, while the shocked path begins from $r_0^1 = r_0^0 - 0.01$ and then follows the same law of motion for interest rates. The market mortgage rate is the model-implied function $m(r_t^z)$.

For each model environment j and interest-rate path z, the cross-sectional distribution evolves according to the Kolmogorov forward equation

$$\frac{\partial g_t^{j,z}(S)}{\partial t} = \mathcal{L}^*_{j,z,t} g_t^{j,z}(S), \qquad g_0^{j,z}(S) = g_0(S),$$

where $\mathcal{L}^*_{j,z,t}$ is the adjoint of the household-state generator. This generator incorporates saving behavior, income transitions, mortgage amortization, exogenous moving, and endogenous refinancing. Endogenous refinancing occurs with intensity

$$\lambda^{refi}_{j,z,t}(S) = \lambda^A_j \left(c - m(r_t^z), s\right) \rho_j \left(r_t^z, S\right),$$

where $\lambda^A_j(\cdot)$ is the attention-arrival intensity and $\rho_j(r_t^z, S)$ is the refinancing execution rule implied by the household problem. Thus, the interest-rate shock affects the transition path by changing rate gaps and the household policy functions evaluated along the rate path. When refinancing occurs, the coupon is reset to the prevailing market mortgage rate $m(r_t^z)$, and households may also adjust mortgage balances through the cash-out choice.

Aggregate consumption in environment j under path z is

$$\bar{C}^{j,z}(t) = \int C_j(r_t^z, S)\, g_t^{j,z}(S)\, dS,$$

where $C_j(r_t^z, S)$ is the household consumption policy. The consumption impulse response for environment j is the percentage deviation of aggregate consumption in the shocked path from the unshocked path:

$$IRF_C^j(t) = \frac{\bar{C}^{j,1}(t)}{\bar{C}^{j,0}(t)} - 1.$$

The figures report $100 \times IRF_C^j(t)$.

I summarize the dynamic response using cumulative impulse responses. For horizon T, the undiscounted cumulative response is

$$CIRF_C^j(T) = \int_0^T IRF_C^j(t)\, dt,$$

and the discounted cumulative response is

$$CIRF_{C,\delta}^j(T) = \int_0^T e^{-\delta t} IRF_C^j(t)\, dt,$$

where I set $\delta = 0.14$ in the baseline reporting. Because $IRF_C^j(t)$ is measured as a proportional deviation, $100 \times CIRF_C^j(T)$ is reported in percentage-point-years. The average response over the horizon is $CIRF_C^j(T)/T$.

For counterfactual comparisons, I compare the cumulative response in the counterfactual economy to the cumulative response in the baseline economy:

$$\Delta CIRF_{C,\delta}^{CF}(T) = CIRF_{C,\delta}^{CF}(T) - CIRF_{C,\delta}^{B}(T),$$

and report the amplification relative to the baseline response as

$$Amplification(T) = \frac{CIRF^{CF}_{C,\delta}(T) - CIRF^{B}_{C,\delta}(T)}{CIRF^{B}_{C,\delta}(T)}.$$

In the exercises, this statistic is evaluated at $T = 5$ years.

F Appendix: Robustness for Monetary-Policy Refinancing Responses

F.1 Alternative Reduced-Friction Scenarios

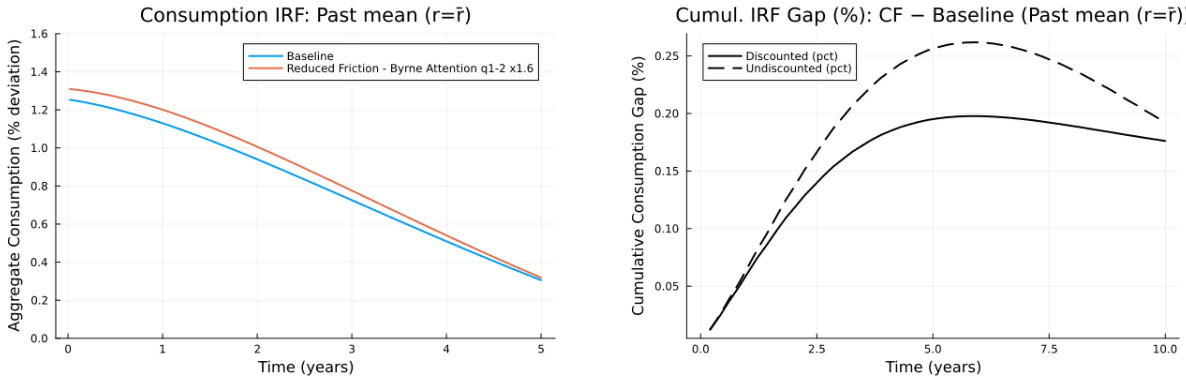
Targeted low-income attention intervention. The first scenario is a targeted version of the Byrne-scaled attention counterfactual. Instead of increasing attention capacity for all borrowers, I raise $\chi_{\max}(s)$ by 60 percent only for the bottom two income quintiles, holding the execution-cost parameters and all other primitives fixed:

$$\chi^{cf}_{\max}(s) = \begin{cases} 1.6\chi_{\max}(s), & s \in \{1,2\}, \\ \chi_{\max}(s), & s \in \{3,4,5\}, \end{cases} \qquad \Phi^{cf}(s) = \Phi(s) \quad \forall s.$$

This exercise is more conservative than the market-wide counterfactual in the main text and can be interpreted as a targeted outreach policy aimed at lower-income borrowers. The five-year discounted cumulative consumption response rises from 3.13 percentage points in the baseline economy to 3.33 percentage points, an increase of 0.20 percentage points, or 6.2 percent relative to the baseline cumulative response.

Equalized top-quintile refinancing frictions. The second scenario considers a larger reduction in refinancing frictions. In this counterfactual, all income groups face the same attention and execution frictions as households in the top income quintile:

$$\chi^{cf}_{\max}(s) = \chi_{\max}(5), \qquad \Phi^{cf}(s) = \Phi(5), \qquad s \in \{1, \ldots, 5\}.$$

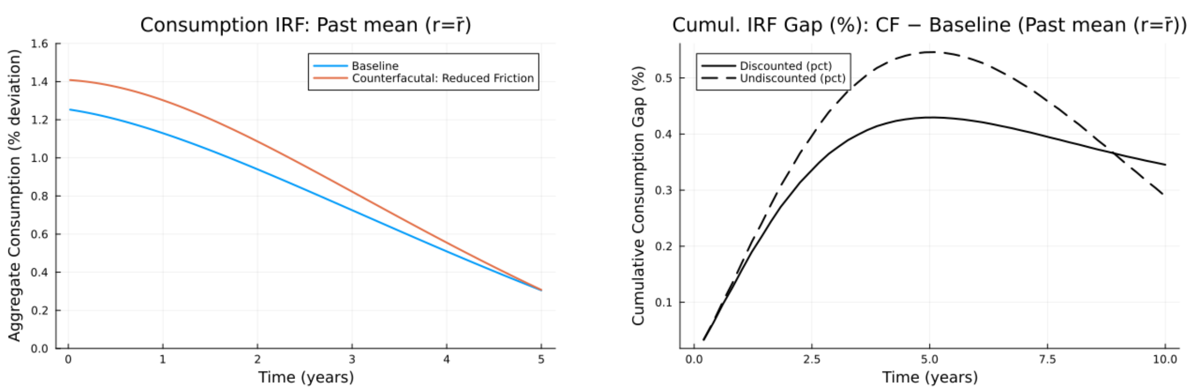


(a) Consumption IRF

(b) Cumulative consumption IRF

Figure F.11: Targeted Byrne-Scaled Attention Counterfactual

All other primitives, the initial distribution, and the interest-rate shock are held fixed. This experiment is not meant to describe a specific reminder policy; rather, it provides an upper-bound-style benchmark for how much monetary transmission would increase if income-related refinancing frictions were compressed to the level faced by high-income borrowers. The five-year discounted cumulative consumption response rises from 3.13 percentage points in the baseline economy to 3.56 percentage points, an increase of 0.43 percentage points, or 13.7 percent relative to the baseline cumulative response.



(a) Consumption IRF

(b) Cumulative consumption IRF

Figure F.12: Equalized Top-Quintile Refinancing Frictions

F.2 MPC Sensitivity

Table F.26 reports sensitivity of the market-wide Byrne-scaled counterfactual to alternative mappings from refinancing-induced payment relief to consumption. The exercise holds fixed the model-implied refinancing and payment-relief paths and varies only the income gradient in MPCs using

$$m_s(\lambda) = \bar{m} + \lambda(m_s - \bar{m}),$$

where m_s is the model-implied average MPC for income state s and $\bar{m}$ is the cross-sectional average. The structural model implies a 10.0 percent discounted amplification of the five-year cumulative consumption response. In the accounting exercises, the discounted amplification ranges from 3.4 to 4.1 percent. The amplification therefore remains positive even when the MPC-income gradient is flattened substantially.

Table F.26: MPC Sensitivity: Market-Wide Byrne-Scaled Attention Counterfactual

Specification	Baseline IRF	Counterfactual IRF	Amplification (%)
Structural model	3.133	3.445	9.959
Accounting, $\lambda = 0$	0.403	0.420	4.130
Accounting, $\lambda = 0.5$	0.408	0.424	3.885
Accounting, $\lambda = 1$	0.413	0.428	3.645
Accounting, $\lambda = 1.5$	0.417	0.431	3.411

Notes: The table reports discounted five-year cumulative responses. The structural model row corresponds to the main counterfactual in the text. The accounting rows hold fixed the model-implied payment-relief paths and vary only the MPC schedule $m_s(\lambda) = \bar{m} + \lambda(m_s - \bar{m})$.